

AI, Political Opinion, and Behavior

How AI Automation of Jobs Influences Political Opinions and Behavior

Master's Thesis

M.Sc. European Governance
M.A. Politics and Administration

by

Simon G. Duris

Student ID: 2157063

Supervised by:
Prof. Rose Vincent
Dr. Tobias Tober

Faculty of Law, Economics and Governance
Utrecht University
Utrecht, Netherlands

18th July 2025

Abstract

Technological innovation, from agricultural mechanization to computerization, has transformed economies in the past. Artificial intelligence (AI) represents the next wave of workplace automation, yet its effects on political preferences and voting behavior have remained unstudied so far. With AI predicted to affect most modern jobs, understanding who will be affected and how this influences political behavior is crucial. I develop an economic framework predicting AI automation winners (high-skill, high AI exposure) and losers (low-skill, high AI exposure). I argue that losers will favor more redistribution and oppose immigration, with the former mediating populist left voting and the latter mediating populist right voting. Using ESS data (2015-2024) with matching and mediation analysis, I find AI losers indeed favor more redistribution and less immigration, but only immigration attitudes translate into voting behavior, specifically for populist right parties. Notably, anti-immigration attitudes also predict populist left voting more than redistribution preferences. All in all, results demonstrate that (AI) automation losers misattribute economic hardship, changing political preferences and behavior in response to AI's negative effects. This cautions against an AI transformation without policies supporting AI automation losers.

Contents

1	Introduction	4
2	Literature Review and Theory	6
2.1	Technological Change and AI	6
2.1.1	Labor Market Effects of AI	8
2.1.2	Losers and Winners of AI-Induced Change	9
2.2	Effects of (AI) Automation on Political Opinions	11
2.2.1	Effect of AI on Redistribution Preferences	13
2.2.2	Effect of AI on Immigration Attitudes	14
2.3	AI and Voting	15
3	Research Design	17
3.1	Independent Variable	17
3.2	Dependent Variables	20
3.3	Control Variables and Matching Covariates	21
3.4	Model Specification	24
3.4.1	Initial Specification	24
3.4.2	Matching	25
3.4.3	Analysis on Matched Dataset	27
3.4.4	Mediation Analysis	29
4	Results	30
4.1	Descriptive Analysis	30
4.2	Inferential Analysis	33
4.3	Mediation Analysis	37
5	Conclusion	39
5.1	Limitations	39
5.2	Discussion	40
A	Appendix 1	48
B	Appendix 2	54
C	Appendix 3	57

List of Figures

1	The relationship between AI Exposure, Political Opinions, and Behavior	17
2	The relationship between AI Exposure and Skill Level	19
3	Density of Propensity Scores in Treatment and Control Group Before and After Matching . .	27
4	Love Plot of Pre- and Post-Matching Balance	28
5	Distribution of AI Exposure Score across ISCED Education Levels	31
6	Map of Mean AI Exposure and AI Vulnerability	32

7	Comparison of Scenarios using OLS results based on the matched data	36
8	Mediation Pathways from Automation Vulnerability to Populist Voting	38

List of Tables

1	Descriptive Statistics of Key Variables	22
2	Linear Mixed Effects Models (Unmatched Sample, Random Effects)	34
3	Treatment Effects of Automation Risk on Political Attitudes (Matched Sample)	35
4	Average Treatment Effects of Automation Risk (Matched Sample)	37
5	Mediation Analysis: Pathways from Automation Vulnerability to Populist Voting	37
A1	Party Classification by Country	48
A2	Variables from European Social Survey (ESS)	52
A3	Propensity Score Matching Specifications and Performance	54
A4	Complete Covariate Balance Diagnostics for Ultra-Tight Matching Specification	55
A5	Mediation Sensitivity Analysis: Thresholds for Eliminating Effects	56
A6	Effect of Automation Risk on Political Attitudes (Unmatched Sample, Fixed Effects)	57
A7	Linear Mixed Effects Models (Matched Sample, Random Effects)	58
A8	Treatment Effect Stability Across ChatGPT Release	58

1 Introduction

Artificial intelligence (AI) is transforming the world of work in unprecedented ways. Unlike previous technological changes that mainly affected routine jobs in factories and offices, AI can perform complex, non-routine tasks that were once thought to be unautomatable. Medical diagnosis, legal research, financial analysis, and, to some extent, creative writing are increasingly within AI’s capabilities. Yet, the phenomenon of AI is not as recent as it might seem. Since 2015, there has been a steep growth in AI-based patents, with the 2023 breakthroughs in large language models (LLMs) accelerating this trend exponentially and bringing it to the attention of the mainstream. These breakthroughs, which took the form of chatbots, such as OpenAI’s ChatGPT or Anthropic’s Claude, spurred an interesting public debate. On the one hand, you have optimistic stories about how AI will spur worldwide economic growth while allowing us to work less and live a life of leisure. On the other hand, there are the pessimistic reports that predict a ‘bloodbath’ of jobs, where many people will face economic difficulties and lose their jobs and income. While these two positions are wildly different, they share the same core: they both assume wide-ranging economic changes. That AI has the potential to change an unprecedented number of jobs in our economies has been shown by previous studies (Eloundou et al., 2024; Fenoaltea et al., 2024; Webb, 2020), but who will be affected in which way has not been examined extensively. And the political and societal consequences of such changes have also only received limited attention.

Economic disruption and political upheaval often go hand in hand. Previous research has shown that workers displaced by previous waves of automation and globalization are more likely to support populist parties and anti-establishment movements (Anelli et al., 2019; Milner, 2021). Brexit, Trump’s election, and the rise of populist parties across Europe all coincided with major economic transformations that resulted in many workers feeling left behind. Yet we know surprisingly little about how AI automation, even though such major economic changes are predicted in its wake, can affect political attitudes and voting behavior. This gap in knowledge is problematic because AI differs fundamentally from previous technological changes and is likely to affect different workers and employees than previous waves, creating new patterns of winners and losers that cut across the established lines.

Thus, this study asks a straightforward but important question: ***How does AI automation affect workers’ political opinions and behavior?*** One key insight I develop is that AI’s impact depends not just on whether someone’s job is exposed to AI automation, but on whether they have the skills to benefit from working alongside AI or face the risk of being displaced by it. A software engineer whose productivity increases with AI coding assistants faces a very different future than a data entry clerk whose entire job could be automated. By distinguishing between AI’s winners and losers, this research examines how AI automation vulnerability shapes support for redistribution, attitudes toward immigration, and voting. Using data from across Europe, I provide the first comprehensive analysis of how AI-driven economic change translates into political preferences and behavior.

This paper makes several important contributions to the academic literature on technology and politics. First, while existing studies have examined how automation affects political behavior, they have focused primarily on robotics and the displacement of jobs with routine tasks, which mainly threatened blue-collar workers (Anelli et al., 2021; Gallego and Kurer, 2022; Milner, 2021). AI automation is fundamentally different from these waves of automation because it can affect both high-skilled and low-skilled workers, but in different ways. In a first effort to systematically distinguish between AI winners and losers, I do not just look at simple measures of technological exposure that treat all affected workers the same, but examine how the impact of AI exposure might differ depending on workers’ skill level. Second, the paper addresses a key theoretical gap in

understanding the causal mechanisms linking automation to political behavior. As Gallego and Kurer (2022), in their recent literature review highlighted, we know that automation-vulnerable workers are more likely to vote for populist parties, but we lack a clear understanding of whether this happens because of a direct link or whether they misattribute their economic hardship through changed attitudes about redistribution, immigration, or other pathways. By testing mediation mechanisms, I provide crucial evidence about how economic transformation translates into political preferences and ultimately behavior.

Due to AI becoming likely very widespread, understanding its political consequences is essential for democratic governance and policy design. As AI adoption accelerates across most sectors of the economy, political leaders need to anticipate how these changes will affect public opinion and electoral outcomes. The paper helps explain why specific groups might be drawn to populist parties as AI adoption increases, providing early warning signs for the potential of political disruption. From a policy perspective, the research offers insights into which workers are most vulnerable to AI-driven automation and what types of interventions they might favor. For example, if redistribution preferences drive support for left populist parties among AI losers, then social insurance policies might help stabilize political support. If immigration attitudes are the key pathway to right populist voting, then different policy responses would be needed. By identifying who becomes politically mobilized by AI and through what mechanisms, this research can inform more targeted and effective policy responses to technological change.

My central theoretical argument is, that AI automation creates distinct winners and losers based on workers' skill levels and their exposure to AI in their occupation. Workers with high skills in highly AI-exposed jobs are positioned to benefit from AI complementarity as it augments their productivity and creates new opportunities for them. In contrast, workers with low skills in highly AI-exposed occupations face the risk of displacement since AI can potentially automate their entire jobs rather than just assist them. This fundamental difference in how AI affects workers should translate into different political preferences. Automation losers, facing economic insecurity and potential job and income loss, should favor more government redistribution to insure themselves against future income loss. They should also develop more negative attitudes toward immigration due to feelings of economic competition and status threat. These changed attitudes should then drive voting behavior, with redistribution preferences leading automation losers toward populist left parties that promise economic protection, while anti-immigration attitudes push them toward populist right parties that blame outsiders for economic problems.

The empirical analysis strongly supports the core theoretical predictions using matched data from the European Social Survey (ESS). Workers facing negative AI exposure (low-skilled, high AI exposure) show significantly higher support for government redistribution and more negative attitudes toward immigration compared to AI winners (high-skilled, high AI exposure). However, the mediation analysis reveals an asymmetric pattern in how these attitudes translate into voting behavior. Immigration attitudes serve as the primary pathway linking automation vulnerability to populist voting, mediating approximately 37% of the effect on populist right party support. In contrast, redistribution preferences account for only 7% of the effect on populist left voting. Interestingly, there is also a cross-pathway where anti-immigration attitudes positively predict populist left voting, though to a lesser extent than populist right voting. Whether this is explained by populist left parties moving towards more anti-immigration attitudes or if this is an expression of a more general anti-establishment sentiment remains an open question. Nonetheless, these findings indicate that immigration attitudes, rather than redistribution preferences, are the more politically consequential outcome of AI-driven economic insecurity.

These theoretical and empirical explanations make three key contributions to our understanding of techno-

logical change and political behavior. First, it provides the first systematic exploration of how AI automation specifically affects political attitudes and voting behavior, moving beyond previous studies that focused on robotics and routine job displacement. Second, the study offers a novel framework for understanding AI’s distributional effects by distinguishing between winners and losers based on the interplay between exposure and skill rather than simply treating all AI-exposed workers as uniformly threatened, revealing that the same technology can simultaneously benefit and harm different groups of workers. Third, the findings provide strong empirical support for the misattribution hypothesis (Gallego and Kurer, 2022), the idea that workers experiencing automation-related economic insecurity redirect their frustrations toward immigration rather than correctly identifying technology as the source of their economic hardship. The fact that immigration attitudes, rather than redistribution preferences, serve as the primary mediating pathway suggests that automation losers are indeed channeling their economic anxieties toward culturally salient out-groups rather than demanding policies that directly address technological displacement.

The remainder of this paper proceeds as follows. Section 2 reviews the existing literature on technological change and political behavior, develops the theoretical framework distinguishing AI winners from losers, and presents the specific hypotheses linking automation vulnerability to redistribution preferences, immigration attitudes, and populist voting. Section 3 outlines the research design, including the construction of the AI exposure measure using Webb’s (2020) patent-based index, the European Social Survey data covering 30 countries from 2015-2024, and the propensity score matching strategy, as well as the mediation analysis. Section 4 presents the empirical results. The final section discusses the limitations of this research and the broader implications of these findings for understanding technological change and democratic politics in the age of artificial intelligence.

2 Literature Review and Theory

2.1 Technological Change and AI

Technological change, historically speaking, is one of the premier drivers of economic growth and prosperity. An easy example that comes to mind is the invention and adoption of steam power and electricity, which led to the mechanization of agriculture. While this destroyed the majority of jobs in farming, it dramatically increased the productivity of labor in manufacturing. Later, automation in manufacturing, based on computer techniques, led to the automation of precision production jobs, while the newly available digital data made analytical and managerial skills particularly valuable to workers and employees. Yet, while technological breakthroughs often happen very suddenly, their adoption in the economy is much more gradual. Regarding the first example from above, the share of agriculture workers in the US labor market didn’t drop overnight; it steadily declined between 1880 and 1970 by around 20% per decade (Deming et al., 2024).

From an economic perspective (for example: Acemoglu and Restrepo, 2019), automation is the development and adoption of new technologies that lead to the substitution of labor in a range of tasks previously performed by labor with capital. This displacement effect changes the task content of production adversely for labor. Thus, the share that labor adds to the produced value reduces due to automation. There are not just historical examples of automation, though. In the last 25 years, we have seen the rise of industrial robots and other automated machinery that have displaced production workers, while specialized software and artificial intelligence take over more and more tasks previously performed by white-collar workers (Acemoglu and Restrepo, 2019; Graetz and Michaels, 2018).

Artificial intelligence and machine learning (ML), the terms are used interchangeably here, are arguably

the most promising technological developments in recent memory, with potential applications in many industries and functions (Alekseeva et al., 2021). While the field of AI research is not just limited to ML, where developers construct complex statistical algorithms that can learn and generalize from data, the most successful commercial innovations are concentrated in this field. Well-known large language models, such as ChatGPT, are based on a branch of ML research that is called deep learning, which allows the representation of texts, images, audio, video, and other data as multilayered neural networks. In a sense, AI and ML technologies are a downstream innovation of the information technology revolution that began in the 1970s and 80s, since they still are technologies for analyzing, interpreting, and responding to digital data (Deming et al., 2024). What differentiates AI/ML from previous software innovations is the way in which data is processed. While conventional software works in a top-down fashion, where explicit instructions are scripted in advance and then routinely and rigorously followed, AI and ML algorithms work from the bottom up. They 'learn' about their environment by studying and copying the actions and implicit rules they infer from patterns in the vast data they are trained on (Autor, 2015; Brynjolfsson et al., 2018). It can be argued that AI is best understood as a prediction technology (Agrawal et al., 2019). It can predict the likelihood that a medical image indicates a pathology, the next words in standard office documents and emails, and the legal liability based on contract language, with a lot more possibilities. Almost all occupations require some tasks that can be thought of as prediction, meaning AI has the potential to impact aspects of nearly all jobs (Eloundou et al., 2024).

It has been argued that AI should be understood in terms of general-purpose technologies (GPTs). The idea of GPTs stems from the study of key technologies that played a central role in fostering economic growth in the past. For a technology to constitute a GPT, it needs to fulfill three criteria: It must be widely used, it must be capable of ongoing improvement, and it must enable complementary innovations across different application sectors (Bresnahan, 2010). Past examples of GPTs include the aforementioned steam power and electricity, but also process and management innovations via the storing and encoding of digital innovation or the Toyota Lean Management (TLM) ¹ approach to manufacturing (Deming et al., 2024). Considering the definition of GPTs and the definition of AI technology outlined above, it is not surprising that AI can be considered a significant general-purpose technology (Brynjolfsson and McAfee, 2017).

Looking at the historical evidence, GPTs lead to productivity growth that follows a 'J curve' where growth is at first lagging because new investments are required to complement the introduction of GPTs (Brynjolfsson et al., 2021). Once those are made however, the productivity growth enabled by GPTs creates new tasks and occupations. Autor et al. (2024) are able to show that from the 1940s onward, technological progress has fully replaced certain jobs and complemented others, leading to the creation of 'new work'. Jobs that did not exist before the invention of certain technologies made them feasible and necessary. Importantly, they differentiate between those innovations that augment labor by directly increasing its productivity and those innovations that automate labor and lead to employment decline.

This latter aspect is in line with the idea that technology has a dual effect. On the one hand, it replaces human labor through automation (Acemoglu and Restrepo, 2019), on the other, it complements human labor by making it more productive and creating new types of work (Autor et al., 2024). This is evident in the fact that when AI innovations' impact on labor is discussed in popular media there seem to be only two possible answers. Either it is seen as the precursor to widespread joblessness and mass unemployment or it is argued to lead to unprecedented economic growth and thus to increasing labor demand and prosperity for all (Acemoglu and Restrepo, 2019).

¹TLM is a socio-technical system of manufacturing developed by Toyota, and builds upon the ideas of just-in-time and jidoka. The idea of 'lean management' stems from TLM

I will discuss this dual nature of (AI) technologies’ impact on the labor market in more detail in the next section, but want to highlight a fact that receives almost unilateral support in the literature. Namely, AI will have a widespread impact on almost all jobs. Brynjolfsson et al. (2018) use their occupation level measure of the suitability for machine learning of individual tasks to show that ML can automate and augment tasks across a broad range of low to high-paying jobs. Similarly, Eloundou et al. (2024) argue that almost 80 percent of US workers have jobs with tasks of which at least 10 percent could be automated by AI applications. Demand for AI skills has grown by a factor of 4 between 2010 and 2019, with this growth accelerating in the last three years of the period and demand still gaining momentum (Aleksieva et al., 2021). This growth is not just limited to IT-related industries and jobs but also occurs in other sectors and occupations.

2.1.1 Labor Market Effects of AI

As mentioned above, AI is arguably a GPT similar to other disruptive innovations. But, at the same time, it is likely still too early to assess its full impact on labor markets. Nonetheless, AI is predicted to impact a broader range of occupations than previous and recent waves of automation have. Not just blue-collar production jobs, which were and are being automated with robotics, but also white-collar occupations that center around non-routine and cognitive tasks (Borwein et al., 2025). Yet, this false dichotomy between alarmist warnings of the end of work by humans and dismissal of concern about labor market disruptions caused by AI automation is counteracted by the early data that is available about labor market changes (Acemoglu and Restrepo, 2018; Deming et al., 2024).

The previous wave of automation led to a polarization of labor markets in the first two decades of the 2000s. Middle-skilled (usually unionized) production jobs were replaced by service sector jobs, which offered lower pay and less employment protection (Autor and Dorn, 2013). But since 2016, low- and middle-skilled jobs have declined, and service sector jobs are now at a similar level to what they were 20 years ago. Thus, they have given back the rapid job growth they experienced. Similarly, retail sales jobs have also declined over the last decade. Conversely, the growth in high-skill occupations has continued, particularly in management, business, education, and healthcare jobs. And while STEM (science, technology, engineering, math) jobs declined in the early 2000s, they are now growing rapidly (Deming et al., 2024).

This is more evidence of the aforementioned dual nature of technologies’ impact on the labor market, also known as the *displacement* and *reinstatement* effects (Acemoglu and Restrepo, 2019). Quite obviously, historical technological innovations did not just displace human labor via automation. If this were the case, we would see a decreasing number of tasks and jobs, but what has happened is that this displacement effect of automation has been countered by technology generating new tasks in which labor has a comparative advantage. Acemoglu and Restrepo (2018) argued that the displacement effect can reduce the demand for labor, wages, and even employment. Additionally, this displacement means that increases in output per worker due to automation do not result in a proportional expansion of the demand for labor. Thus, this displacement effect “causes a decoupling of wages and output per worker, and a decline in the share of labor in national income” (Acemoglu and Restrepo, 2018, p.198). A countervailing force is the fact that the substitution of cheap machines for human labor creates a productivity effect. Because the cost of producing declines due to the automation of tasks, the economy will expand and increase the demand for labor in non-automated tasks. This can happen in automating sectors but also in non-automating sectors (Acemoglu and Restrepo, 2018). More importantly, technology also leads to the creation of new tasks, functions, and occupations in which labor has a comparative advantage over capital. This increases the demand for labor

and increases its share in national income. As Acemoglu and Restrepo (2018, p.206) put it: "[A] powerful way in which technological progress could be associated with a balanced growth path is via the balancing of the impacts of automation by the creation of new tasks".

There are numerous historical examples of this reinstatement effect via the creation of new tasks. During the decades of agricultural mechanization, new tasks in new industries, especially in factories, played a large role in stabilizing employment (Olmstead and Rhode, 2001). In the 19th century, technological innovations led to the inception of entirely new occupations, such as line engineers, machinists, repairmen, conductors, and managers (Mokyr, 1990). The latest digital revolution might have already replaced some white-collar tasks, but has simultaneously created new tasks in relation to programming, design, software and app development, database design and analysis, and computer security. Additionally, it has created tasks with more specialized functions within existing jobs, such as administrative assistants and medical equipment technicians (Lin, 2011). Looking at AI and its application to education or healthcare, it is conceivable that it may result in new employment opportunities. In education, it might be feasible to become more customized. Something that has been shown to benefit students, but was too costly with current technology. AI applications may be able to create more jobs for professionals to design, implement, and monitor individual education programs, with similar prospects existing in healthcare and elderly care services (Acemoglu and Restrepo, 2018). There is a substantial literature that examines the potential of human-AI collaboration and how this could improve performance and productivity (see, for example: Hemmer et al., 2024).

Considering the above paragraphs, the net impact of AI on employment depends upon the balance of the displacement effect and the creation of new tasks and jobs via the reinstatement effect. To put it differently, if the productivity gains from AI increase the size of the pie sufficiently, workers will still get a larger slice than they did before. Even if some slices of the cake are eaten by robots. Yet, it is unlikely that all workers will benefit equally from the introduction of technological innovation. Acemoglu and Restrepo (2022) can show that up to 70% of changes in the US wage structure over the last four decades were driven by automation. To be precise, automation evidently led to the reduction of real wages for workers in routine task industries, such as blue-collar jobs in manufacturing, while those worker groups that were not displaced from their tasks, such as those with a post-graduate degree or college-educated women, saw wage gains. This already makes it clear that automation has a heterogeneous impact on workers, as it produces winners and losers. In the next section, I develop an argument about which workers stand to benefit and which stand to lose from AI-based automation.

2.1.2 Losers and Winners of AI-Induced Change

Often, when looking at the factors of production, labor is typically treated as a homogeneous block. This is, of course, not true. Individual workers differ on countless dimensions. If, as discussed above, technological innovation displaces some jobs and creates other, new jobs, not all workers are likely to benefit to the same degree. Automation and displacement will affect a specific set of tasks and thus a certain set of workers, who bear most of the cost of reduced demand for their work, and thus reduced wages. Conversely, automation will reinstate a certain set of new tasks, a set that specific groups of workers will be able to perform, not all workers. These groups stand to reap most of the benefits, as they will face greater demand and higher productivity, leading to higher wages. Thus, automation and technological innovation have distributional consequences for labor, producing winners and losers. This pattern has been observable in the past and has been dubbed labor market polarization or 'hollowed out middle' (Autor and Dorn, 2013). Recent automation has been focused on routine jobs, mainly in manufacturing and clerical work, and the displacement effect

mainly hit workers at the middle of the skill distribution (Acemoglu and Restrepo, 2019; Borwein et al., 2025). At the same time, newly created jobs either benefited highly educated professionals with cognitive or interpersonal tasks or were created in the low-skilled, manual services sector (Schöll and Kurer, 2024). Thus, losers of past automation were middle-skilled workers, and winners were higher- (and to a lesser extent lower-) skilled workers.

However, AI automation departs from previous technological disruptions to some extent. As detailed above, past automation primarily affected routine workers in the middle of the skill distribution, but AI’s capabilities extend beyond this traditional boundary (Borwein et al., 2025). The recent advances in AI research and their application hint at a wider automation of tasks, and not just those with a routine nature. This makes non-routine work increasingly vulnerable to technological disruption. It will not be just low-skilled / low-wage workers that will be exposed to AI, but also high-skill / high-wage workers (Brynjolfsson et al., 2018; Eloundou et al., 2024; Frey and Osborne, 2017; Webb, 2020). But to what extent these workers will be displaced or reinstated is, to some extent, an open question.

I argue that despite AI’s broad reach, highly skilled workers are positioned to benefit from AI for several reasons, which I will outline here. First, AI creates complementarity opportunities that enhance rather than replace high-skilled work (Hemmer et al., 2024). Task complementarity between AI and human workers can increase productivity in those tasks that are not fully automated. Workers performing these tasks may have better information to access or other complementary inputs they can benefit from. Alternatively, AI may only automate some subtasks, which would enable workers to specialize in the non-automated subtasks and raise their expertise and productivity in those. For example, programmers may be able to outsource the writing of standard subroutines to AI tools, so they can focus on other, more complex, parts of programming (Acemoglu, 2024).

Second, the complexity and high-stakes nature of many high-skilled occupations create barriers to full automation. Fenoaltea et al. (2024) compare the technological feasibility of occupations being automated with AI to actual applications of AI being developed currently. They find that jobs requiring a master’s degree or higher often have high theoretical AI exposure but are not (currently) the target of automating AI applications that are being developed. This highlights that even when there is theoretical potential for AI involvement in these jobs, the necessity of advanced skills, complex task sets, and the necessity to be able to handle high levels of uncertainty, the practical integration of AI is less straightforward (Fenoaltea et al., 2024).

Third, one can distinguish between tasks that are easy to learn for AI and those that are hard to learn for AI. Tasks that are hard to learn for AI also often require a high level of skill for workers to perform. Easy tasks are those that have a reliable, observable outcome metric and where there is a simple mapping between action and the outcome metric (Acemoglu, 2024). An example is the verification of the identity of someone who has been locked out of a security system. Hard tasks are those tasks that do not have a simple mapping between action and outcome. This means, to get to a desired outcome is not obvious beforehand and depends on contextual factors, or there may be many relevant contexts to consider. On top of that, it is often unclear for the AI system what exactly needs to be learned to solve such hard tasks, and the information to do so might also be limited (Acemoglu, 2024). An example is the diagnosis of the cause of a persistent cough and the proposal of a treatment. While this is already hard for humans, it is exponentially harder for current AI systems. Crucially, this distinction differs from that of routine vs. non-routine tasks. Routine tasks have already been automated to a large extent in recent waves of automation, whereas AI promises to automate even non-routine tasks is what makes it stand out (Susskind, 2020). Arguably, tasks

that require highly skilled workers predominantly fall into the category of tasks that are hard to learn for AI. High-skilled occupations are characterized by their requirement to handle uncertainty and make context-dependent decisions, which are precisely the characteristics that define hard tasks for AI (Fenoaltea et al., 2024).

Fourth, domain expertise has been shown to become increasingly valuable in AI-integrated environments. Choudhury et al. (2020) show that workers who possess domain expertise, i.e. the skills and knowledge they have accumulated through prior learning within a domain, are particularly complementary to AI systems. Similarly, vintage-specific skills, i.e. the skills accumulated through prior work with the technology, are more productive. Especially the former point is relevant to my argument since even among high-skilled workers, not many will have worked with AI systems before. Nonetheless, domain expertise increases complementarity because it brings in relevant outside information that can mitigate AI bias that stems from incomplete information in the training data. In line with these arguments, it has been shown that AI technology does increase the demand for high-skilled jobs complementing it. Firms that hire AI-skilled labor, indicating that they make use of AI automation, also increase their demand for high-skilled non-AI labor (Alekseeva et al., 2021).

There is another point one needs to consider that is not directly related to the tasks that high-skilled workers perform. It is the workers in the upper-income decile who also have the lowest share of their total income from wages. They have the largest wealth holdings, and a large share of their wealth is in assets such as firm equity. Thus, they are not only insured against negative labor market impacts but also stand to benefit from an AI-driven rise in capital returns (Rockall et al., 2025). Thus, they will arguably benefit twofold: Their AI complementarity will increase the demand for them, increasing their wages. And second, they will be able to accumulate even more wealth as their investments benefit from AI-driven economic growth.

In contrast to high-skilled workers, those in the middle and lower ends of the skill distribution face more severe displacement risks from AI automation. For the same reasons as outlined above, it is them who will bear most of the displacement effects caused by AI automation. They predominantly perform occupations that are not just partly automatable but rather fully automatable, meaning they do not benefit from any complementarity effects but rather will be displaced from their work. Additionally, middle-skilled workers particularly face the pressure of the ongoing automation of routine (cognitive) work. There is some evidence of occupational restructuring in line with these arguments. Front-office jobs like secretaries and administrative assistants, and back-office jobs like billing and financial processing have declined by more than 50 percent in the US economy since 1990. But at the same time, managers and business analysis jobs have seen steady growth. This seems to capture the trend away from routine monitoring work toward strategy, analysis, and decision-making as highly valuable skills (Deming et al., 2024). In line with this, we see a corresponding difference in subjective risk perceptions. Kurer and Häusermann (2022) can show that higher-skilled workers feel significantly less threatened by automation than lower-skilled workers.

2.2 Effects of (AI) Automation on Political Opinions

Having discussed the substantial effects technological change, and in particular AI-driven automation, has and will have on workers, I now go on to argue how AI automation can impact political opinions.

From a basic theoretical perspective, the idea for this link between structural economic transformations, such as automation, to political opinion formation and voting is relatively straightforward (Anelli et al., 2021). Economic pressures, and in particular technological disruptions, create aggregate gains but also winners and

losers, as discussed above. These distributional consequences then lead to perceived economic distress, which is then turned into politically consequential changes in opinion and behavior. While this has been examined for prior waves of automation, there exists no such research for AI-driven automation to my knowledge. Additionally, the mechanism through which automation leads to changes in political behavior, namely by affecting political opinions, has also only received limited attention in the literature (Gallego and Kurer, 2022).

There is an ongoing debate in the literature about whether the relationship between workplace exposure to automation and voting behavior is a direct or indirect one. It makes sense to assume that the objective automation exposure changes workers' political opinions and behavior because they feel this exposure also as a subjective threat. Yet, research has shown that there exists a decoupling between objective exposure and subjective perceptions (Borwein et al., 2025). Nonetheless, it is arguably possible that objective automation exposure influences political policy and political preferences without being reflected in subjective assessment. Workers in jobs negatively exposed to automation might still perceive their jobs as being in decline, but they might attribute this to other reasons than technological change. Since they will feel like and experience that the demand for the tasks they can perform is in decline, these perceptions, without being tied to an understanding of technological disruption, can influence political preferences. Meaning other factors, such as globalization, are blamed as the culprit for adverse labor market changes (Gallego and Kurer, 2022).

Workers experiencing automation-related economic insecurity are unlikely to correctly identify technology as the source of their challenges, leading to systematic misattribution of technology-driven job displacement to more politically salient targets such as immigration, trade, and offshoring. This misattribution occurs partly because automation-related job transformation has been less politicized than other sources of job loss (Gallego et al., 2022), and also because politicians actively redirect blame away from technology toward global forces and demographic changes (Wu, 2022). Technology presents an unappealing political target since the public generally favors technological progress, making policies to slow technological change politically unfeasible, whereas blaming global forces offers ostensibly simpler policy solutions like restricting trade and reducing immigration. Consequently, politicians have effectively decoupled objective and subjective exposure to automation in voters' minds, with the result that objective technology exposure triggers changing political preferences not through direct concerns about technological displacement, but through workers' misdirected blame toward other economic and cultural forces (Borwein et al., 2025).

This redirection of blame is made possible by the fact that automation is one of many labor market risks. The general risk of unemployment for an individual worker is made up of a number of factors, of which automation is only one. Other mechanisms that might threaten one's employment or at least one's wages are: globalization, via the offshoring of labor abroad; consumer preference changes, where demand for certain products disappears; rationalization, meaning a reduction of workers to increase the profit margin; etc... (Im et al., 2019).

Recent research has been able to show that those workers who are directly threatened by a transforming employment structure do seek to express their discontent (Anelli et al., 2019; Frey and Osborne, 2017; Im et al., 2019; Milner, 2021). This is particularly unsurprising because social policy responses to labor market restructuring have been limited by the fact that governments are constrained by fiscal pressures, which forces governments to prioritize their social policy spending (Busemeyer and Tober, 2023).

Another way in which automation can change workers' political preferences and behavior is the idea of occupational preference formation (Schöll and Kurer, 2024). In short, one can differentiate between technical, bureaucratic, and interpersonal work logics, the setting of the work process, the relation to authority, the

type of client interaction, and the kind of skill applied. The theory posits that occupations with lower education levels, with strict hierarchies, and that deal with objects and files (not people) are related to more authoritarian values. Jobs that are based on cooperation (not hierarchy), which focus on social interaction and culture, and require a university education are associated with more cosmopolitan and progressive values (Schöll and Kurer, 2024). Considering the previous discussion of how AI automation will impact workers, and which workers stand to benefit and which to lose, it becomes obvious how winners and losers of AI-driven automation might differ in their political views.

2.2.1 Effect of AI on Redistribution Preferences

In this section, I discuss how negative AI exposure can influence a particular policy opinion: redistribution preferences. The link between actual and perceived labor market risks with increased demand for redistribution via the welfare state and calls for social protection is strongly supported in the literature (Busemeyer and Tober, 2023; Gallego and Kurer, 2022; Walter, 2010). Since workers are, to some level, aware of their higher risk of unemployment or potentially longer unemployment spells, or reductions in real and relative wage growth, regardless of whether they correctly identify the reasons behind this, they support widespread compensation policies that insure against a loss or reduction of income (Gallego et al., 2022). While this has been established for many different types of labor market risk, it is yet to be examined for AI-driven automation.

This builds upon the insurance model of redistribution preferences. The basic theoretical framework for redistribution preferences in political economy is the Meltzer-Richard (MR) model (Meltzer and Richard, 1981). It posits that individual redistribution preferences are a function of material self-interest. It works under the assumption that the preferences of the median voter determine government policy and that the median voter prefers to maximize their current income. If there were no deadweight costs to redistribution, everyone below the mean income would maximize their utility by imposing a 100% tax rate. When there are distortionary costs to taxation, which there are, then the model posits that, when increasing the distance between the median and the mean income, more inequality should be associated with a higher rate of redistribution. This fundamental framework has been extended by the idea that current material self-interest is not necessarily limited to one's present income. Instead, an insurance component is added to redistribution preferences that adds a temporal element. Workers want to insure themselves against uncertain future income levels, so they favor protective welfare state policies when they assume a heightened risk of income and job loss. Since these forms of social protection come in the form of unemployment benefits and social assistance, they are redistributive and thus individuals exposed to increased labor market risk will have a stronger preference for redistributive state efforts (Rehm, 2009; Thewissen and Rueda, 2019).

There is an additional consideration to be made. Workers for whom technological automation translates to imminent employment risk face an interesting trade-off (Busemeyer and Tober, 2023). They will benefit from the short-term gains of compensation that account for imminent income reductions, but might benefit even more from the long-term gains of skill investments that come with active labor market policies, such as additional training. The problem is that compensatory policies address the need for immediate cash transfers, even though limited, very reliably, whereas the success of active labor market policies, even though potentially more valuable, is not as certain (since the probability of re-employment and future income levels remain uncertain). Thus, workers who face labor market risk due to automation have a clear answer to this trade-off: they prefer more direct redistribution over long-term active labor market policies. One might question whether this is an actual trade-off and why states should be unable to supply both. But this is a

different question that I don't aim to answer here. What this makes clear is that people who face AI-driven automation risks will want redistributive policies, and they might additionally want active labor market policies, but if they have to decide, they will decide for the former.

Following this theoretical reasoning, one can thus expect that workers who can be anticipated to be negatively affected by AI-driven automation will have a stronger preference for redistributive policies than workers who benefit from this automation.

***H1:** Losers of AI-driven automation favor more redistribution than winners of AI-driven automation.*

2.2.2 Effect of AI on Immigration Attitudes

At first thought, a connection between negative automation exposure and changes in one's sentiment about immigration might seem puzzling, since these are two different things. Yet, because people who face economic hardship often feel a sense of frustration and injustice, they seek someone to blame (Bauer et al., 2021). Additionally, opportunistic politicians supply convincing anti-immigrant and anti-minority stories to exploit these feelings during economic crises (Voigtländer and Voth, 2015). On top of that, Bursztyn et al. (2022) argue that economic hardship can facilitate scapegoating by lowering the costs of xenophobic rhetoric and behavior. Because crises provide a plausible rationale for intolerant and xenophobic views, it increases the willingness of people with those views to express them since that plausible rationale lowers the expected social sanctions of expressing such views. Thus, economic hardship can foster anti-immigration sentiment not just by changing people's attitudes directly, but also by making them more willing to express such attitudes.

When arguing from a purely economic perspective, people negatively affected by automation may perceive the threatening economic hardship as a tax on their group's material and intangible resources. The negative prospects of automation destabilize people's feelings that their group has economic and cultural security. This will make other groups that compete for the same resources seem more threatening. In particular, automation can cause negative perceptions about immigrants as these constitute a salient minority group that is stereotyped as economic and cultural competition (Gamez-Djokic and Waytz, 2020).

Even from an economic perspective, automation can increase anti-immigration sentiment not just from feelings of economic competition but also due to symbolic threat perceptions that stem from changes to group identity and status. To be more precise, as a first mechanism, automation activates realistic threat perceptions since an important resource (jobs) is increasingly limited, which, combined with the perception of a potentially competitive out-group that competes for this resource, leads to perceived intergroup competition (Butz and Yogeewaran, 2011). That competition then results in negative feelings towards members of the other groups and efforts to limit the other groups' access to those resources (Krosch and Amodio, 2014). Considering the above discussion about how people negatively affected by automation will maybe have to rely on social security benefits to a greater extent, they might also perceive immigrants as competing with them for these benefits and might thus have another reason to have negative perceptions of immigrants, as they would want to limit their access to this resource as well. And as a second mechanism, automation can lead to cultural change, which can also increase feelings of cultural threat from immigrants. Since (AI) automation will likely transform work, this can 'diminish traditional Western values' which emphasize a devotion to work, discipline, and self-reliance. Thus, people may perceive aspects of their identity as vulnerable, which in turn increases the perception that immigrants with different values pose a threat to the dominant culture (Gamez-Djokic and Waytz, 2020).

The mechanism from economic threat to prejudice also builds upon emotions playing a central role (Rhodes-Purdy et al., 2021). Economic hardship and the threat of economic problems generate high levels of anxiety and anger. This generates new or activates existing latent feelings of cultural discontent. Because fear triggers information-seeking, it leads to increased reliance on new information in political decisions and opinion formation. But because this seeking is biased towards threatening information, it has been shown that people become more suspicious of out-groups, and it predisposes people to be more negative towards groups they already dislike. Thus, anxiety makes people more likely to perceive differences in identity as threatening. Moreover, threats trigger a desire to assign blame, and when individuals find groups they can blame, this results in anger. Angry people often have a strong desire to punish those they deem responsible (Brader and Marcus, 2013), especially since they interpret undesirable events as stemming from intentional wrongdoing by others (Petersen, 2010). In line with this, Banks and Valentino (2012) were able to show that anger activates and increases negative feelings toward cultural out-groups such as immigrants.

Taking all of these theoretical mechanisms into account, one can expect workers, who are and will be negatively affected by AI-driven automation, to have a more negative opinion about immigration compared to workers who are benefiting from this automation.

***H2:** Losers of AI-driven automation are more opposed to immigration than winners of AI-driven automation.*

2.3 AI and Voting

While these attitudinal consequences are important in their own right, they become especially significant insofar as they lead to changes in political behavior. If changes in attitudes are inconsequential for actual behavior, then they are noteworthy but not as important. Concerning the losers of AI-driven automation, I argue that their attitudinal changes are, in fact, reflected in changes in behavior. Specifically, I posit that their voting behavior is affected inasmuch as they are more likely to vote for populist parties.

There are multiple reasons for expecting losers of AI-driven automation to shift to populist parties in general. Almost by their definition, these parties are conceived of, and present themselves, as distinct alternatives to traditional mainstream parties. Economic distress has been shown to foster distrust towards political institutions that goes beyond the standard anti-incumbent sentiment and is more broadly directed against the system of liberal democracy and the institutions that represent it. Hence, it is not surprising that populist parties, which present themselves as also being in opposition to these institutions, are a premier option for voters struggling economically, such as automation losers (Algan et al., 2017; Anelli et al., 2019).

Additionally, mainstream parties from the left and right have themselves contributed to their declining appeal. They have converged on both economic and cultural issues over the last decades (Hall and Evans, 2019). Mainstream left parties, such as many social-democratic parties in Europe, have mediated their economic agenda in an effort to capture more economically centrist votes. What plays into this is that redistribution promises have become less credible as governments are constrained by fiscal austerity promises that prohibit them from enacting more redistributive policies (Bandau, 2023). While this has helped the mainstream left to capture the votes of the professional managerial class who were also attracted by the more cosmopolitan stances, it has led to blue-collar voters becoming more and more important as an electorate for populist parties (Keman, 2011; Oskarson and Demker, 2015).

Populist parties on both, the left and right, offer generic appeal to 'take back control' and thus generic promises of protection. Populist left parties focus on economic issues and redistributing income and wealth

from the top 1% of the wealth distribution to the lower percentiles. Populist right parties offer a more generic promise of returning the country to its supposedly better past, before globalization, computers, and immigration had a disruptive impact. While populist left parties are also critical of economic factors such as globalization and trade, populist right parties often add rhetoric that centers traditional family values and households that are empowered by a single, well-paid job (Akkerman, 2015). It is thus not surprising that nostalgia is an important factor in deciding to vote for populist right parties, and to a lesser extent also for populist left voters (Steenvoorden and and Hartevelde, 2018). These appeals to an era of the past are unsurprisingly particularly attractive to voters who feel their social and economic status is threatened or already in decline (Burgoon et al., 2019).

To be more precise, I argue that populist left parties target AI-driven automation losers’ material concerns. They have the anti-establishment advantage that makes them appealing, and combine it with a pro-redistribution platform that is attractive to economically distressed voters. While automation losers have been shown to be more likely to vote for populist left parties in general (Anelli et al., 2021) it speaks particularly to the idea that populist left parties target material concerns that Borwein et al. (2025) were able to show that voters who feel personally threatened by automation are more likely to vote for populist left parties. Thus, I argue that automation losers are more likely to vote for populist left parties compared to automation winners and that this effect is mediated by redistribution preferences.

***H3a:** The effect of negative AI automation exposure on populist left voting is mediated by redistribution preferences.*

Conversely, as left populist parties aim at automation losers’ immediate material concerns, populist right parties tend to address their status concerns (Kurer, 2020). As mentioned above, populist right policy platforms usually target economically declining regions and ethnoracial groups that think of themselves as having dominant claims to national belonging, promising a restoration of standing and a return to bygone eras that were supposedly more prosperous (Borwein et al., 2025). To package these promises, populist right parties employ narratives that denounce elites, immigrants, and other minority groups. Instead of promising worker protection through compensation via increased redistribution, they want to undo labor market changes with nationalist and nativist policies that limit immigration and trade and promise to restore the status of dominant groups by returning the country to its ‘better’ old days (Bonikowski and Stuhler, 2022; Im et al., 2019). Automation losers are mainly pushed to vote for populist right parties due to collective status threats and not necessarily by a fear of immigrants stealing their jobs. While this also may play a role, it is the anxiety and anger over status loss that mainly drives automation losers to develop anti-immigrant sentiment and vote for populist right parties. This is supported by the fact that workers who feel a more collective automation threat are more likely to vote for populist right parties (Borwein et al., 2025). Thus, I argue that automation losers are more likely to vote for populist right parties compared to automation winners and that this effect is mediated by immigration attitudes.

***H3b:** The effect of negative AI automation exposure on populist right voting is mediated by immigration attitudes.*

To visualize the proposed relationships between negative AI automation exposure, Figure 1 employs a flowchart. In the top half, the direct effect of negative AI exposure on redistribution preferences translates into voting for populist left parties, highlighting the mediating role these preferences play. A direct effect of negative automation exposure nonetheless remains and also stands in for the fact that there might be other

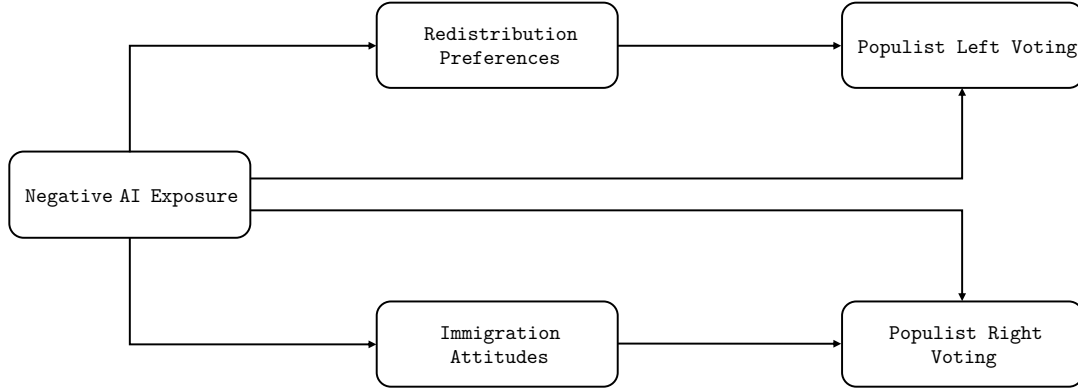


Figure 1: The relationship between AI Exposure, Political Opinions, and Behavior

factors that additionally mediate the effect on voting for populist left parties. Conversely, on the bottom half Figure 1 visualizes how negative AI exposure influences attitudes about immigration and how those translate into votes for populist right parties. Again, there is a further direct effect of negative AI automation exposure that additionally highlights the potential for other factors also mediating the relationship between AI-driven automation losers and populist right voting.

3 Research Design

To test the proposed hypotheses, this paper employs a large-N, quantitative research design. In this section, I detail the conceptualization and operationalization of my variables and the matching and mediation approaches I take.

3.1 Independent Variable

My main argument builds upon the assumption that the development and proliferation of new and better AI technology will have a major impact on the labor market. Since I am interested in the effect AI-imposed job insecurity has on individual workers and employees, I need to gauge how AI influences someone’s job and how their job is, and will be affected by AI. Thus, I need a measure of how much different jobs are and will be impacted by AI, i.e. whether it is or will likely be possible to either fully or at least partially automate them. Classic examples here are translators or interpreters (Eloundou et al., 2024). Conversely, I need to know which jobs will only be affected marginally, or not at all, by AI. A typical example being hairdressers and barbers. These are occupations that AI will not be able to automate since they require physical interaction, and even automation by robots seems far out of reach for now. For this, I first and foremost rely on a measure

by Webb (2020) who uses the overlap between the text of job task descriptions and the text of patents to construct a measure of the exposure of tasks to automation.

The measure is based on the task-based framework of automation developed by Acemoglu and Restrepo (2018). The model assumes that automation occurs at the level of tasks. At the same time, it is also assumed that occupations produce output using tasks, and firms in turn produce output using occupations. This means that when we understand how technology affects tasks, we can also understand how occupations are affected by that technological change. This is central to my argument about the individual-level effects of AI technological development on opinions. To understand how AI affects different tasks, Webb (2020) extracts verb-noun pairs from descriptions of tasks that are associated with all occupations in the *ONET Database, which is produced by an agency of the US Department of Labor. The database was developed to describe occupations and includes natural language descriptions of the tasks that are relevant to it, as well as numerical information on how important each task is to that occupation. Similarly, verb-noun pairs are extracted from the descriptive, natural language titles and descriptions of patents that deal with AI technology. For each pair, and pairs similar to it, it is calculated how often they occur in the full list of pairs. Take the pair 'diagnose, disease' as an example, it might represent 0.1% of all pairs extracted from the set of artificial intelligence patents. To each occupation-based pair, the relative frequency of similar pairs from the patents is assigned. If the pair 'diagnose, condition' and similar pairs are represented in 0.2% of all patent-based pairs, they get assigned a score of 0.002. To then arrive at a single exposure score for each occupation, the average of all verb-noun pairs that were mentioned in the description of that occupation is taken, weighted by the importance of the task to the occupation. The result is a variable that measures the exposure of each occupation to automation by artificial intelligence development.

Several methodological approaches have emerged to quantify occupational exposure to artificial intelligence, each with distinct analytical strengths and limitations. The seminal work by Frey and Osborne (2017) employs expert assessments to estimate automation probabilities for occupations, while Brynjolfsson, Mitchell, and Rock (2018) develop a Suitability for Machine Learning (SML) rubric evaluating task-level automation potential. Similarly, Felten, Raj, and Seamans (2019) construct an AI Occupational Impact (AIOI) measure linking technological progress metrics to occupational abilities, and more recently, Eloundou et al. (2024) introduce an exposure framework using human and LLM-based task evaluations. While these approaches offer valuable insights, Webb's (2020) patent-based methodology provides several analytical advantages, particularly suited to this study's theoretical framework. As artificial intelligence constitutes a general-purpose technology requiring extensive complementary innovations and software developments (Eloundou et al., 2024), patent data captures not merely current AI capabilities but the evolving trajectory of technological development and associated supplemental innovations. This forward-looking dimension is particularly valuable given AI's nascent stage and rapid evolution. Furthermore, the moderate correlation between Webb's measure and alternative AI exposure metrics (correlation coefficients ranging from 0.228 to 0.55 across different studies) suggests methodological robustness while maintaining sufficient independence to capture distinct aspects of AI's occupational impact. This provides a theoretically grounded and empirically reliable foundation for examining the political implications of AI-induced labor market transformation.

As I argued previously, how people are impacted by AI does not only depend on their exposure but also on their skill level. For the reasons outlined above, I assume people in high-skilled jobs who will be strongly impacted by AI will face a type of positive uncertainty. To reiterate, AI will impact the labor market in two interrelated ways: by augmenting and automating jobs (Eloundou et al., 2024). The task model of automation shows that jobs consist of a number of tasks. For some of these tasks, AI is able to perform them

automatically. If enough of the important tasks that constitute an occupation are able to be performed by AI automatically, the job will be automated. All other jobs can only be augmented by AI, where particular tasks may be automated, but a human will still be necessary to do most of the work. Additionally, there are those jobs where AI will not play a role at all. I now argue that high-skilled workers and employees will predominantly work in occupations that are augmentable by AI but are not fully automatable. These jobs often involve critical thinking, as well as cognitive and interactional skills that require tertiary education (Schöll and Kurer, 2024). On the other hand, workers at the lower end and middle of the skill distribution are predicted to be more negatively impacted by AI developments. When they work in jobs that are highly impacted by AI, their jobs tend to be not just augmented by AI but rather automated. Thus, workers and employees who have lower skills and work in occupations that are highly impacted will face a type of negative uncertainty.

Figure 2 illustrates this. The y-axis shows the level of AI exposure of jobs from low to high. The x-axis visualizes the skill distribution from low to high. According to my argument, workers and employees in the upper left quadrant will be highly negatively affected by AI, while people who find themselves in the upper right quadrant will be highly positively affected by AI. Since AI is a general-purpose technology (Eloundou et al., 2024), it will affect almost every job to some extent, and thus, a muted effect on those who are comparatively sheltered from AI exposure is also possible. Parallel to those who work in jobs with high exposure, skill level is a deciding factor. Those with a low skill level, and thus in the lower left quadrant of Figure 2, will also be negatively affected but not as strongly, and thus I expect a muted effect. Conversely, those in the lower right quadrant have a high skill level and should be positively affected, but also in a more muted way.

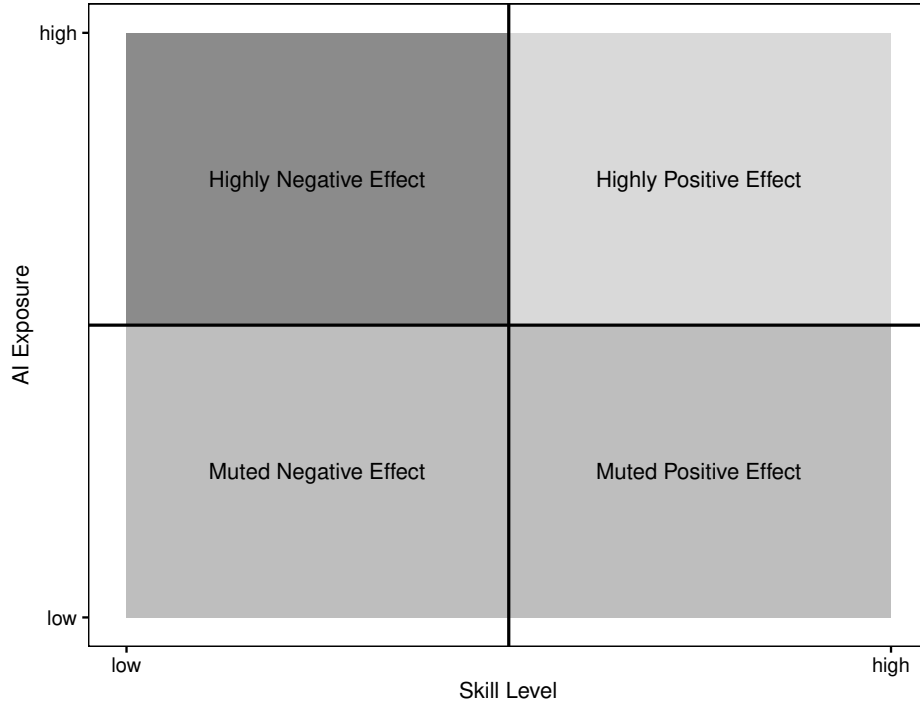


Figure 2: The relationship between AI Exposure and Skill Level

I rely on the European Social Survey (ESS, 2009, 2011, 2013, 2015, 2017, 2019, 2022, 2024) to do my

analysis. The ESS is a cross-national survey that has been released every two years since its inception in 2001. Data is collected via face-to-face interviews; cross-sectional, newly selected samples that are representative of a country’s population aged 15 and older are used. The latest available data was collected in 2023 and 2024 and released in the last year. Importantly, the ESS has collected information on respondents’ occupations, classifying them according to the International Standard Classification of Occupations (ISCO), developed by the International Labor Organization (ILO). The ESS uses the 4-digit codes developed by the ILO, which are the most granular of the 4 levels that are available in the ISCO. I use the official crosswalk from the *ONET data to the ISCO data to merge the ESS data with the automation exposure data by Webb (2020). In doing so, I now know how exposed each respondent’s occupation is to AI. To gauge the skill level of a respondent, I use their education as a proxy. While initial education does not necessarily translate to someone’s skill level, as they might have continued training and education during their working life, the initial education someone received is nonetheless a good proxy for their overall skill level (Walter, 2010). The ESS measures someone’s education by classifying them according to the International Standard Classification of Education (ISCED). The ISCED provides 7 categories that range from only receiving primary education to obtaining a Master’s degree or higher.

Having a measure of someone’s exposure to AI and their skill level, I construct an *AI Score* variable that ranges from 1 to 4. Respondents who have an AI exposure that is higher than the median AI exposure in the sample and fall into the ISCED category 5 or lower, meaning they have not received any tertiary education, receive a score of 4. They are the people who would fall into the upper left quadrant of Figure 2 and constitute the AI automation losers. Respondents with an AI exposure lower than the median exposure level and no tertiary education receive a score of 3. They would lie in the lower left quadrant of the Figure. People with an exposure score that is also lower than the median exposure but have a higher level of education, so fall into an ISCED category higher than 5 and have received some form of tertiary education, lie in the lower right quadrant and receive a score of 2. Lastly, respondents with an AI exposure greater than the median and in an ISCED category of 6 or higher, meaning having received a tertiary education, receive a score of 1 as they fall into the upper right quadrant of Figure 2. They constitute the winners of AI automation.

3.2 Dependent Variables

My first two hypotheses necessitate two dependent variables. Both are also based on the ESS. In all rounds used in this analysis, the respondents were presented with the following statement: “The government should take measures to reduce differences in income levels”. They were then asked to express to what extent they agreed or disagreed with the statement, using 5 categories that range from strongly agree to strongly disagree. This is the basis for my dependent variable that measures *Redistribution Preferences*. I recoded the ESS variable so that a score of 5 indicates strong support for redistribution measures while a score of 1 indicates opposition to such measures.

In a similar vein, the respondents were presented with the statement: “Would you say it is generally bad or good for [country]’s economy that people come to live here from other countries?”. They were able to answer this question by giving a score on a scale from 0 to 10 where 0 means immigrants are bad for the economy and 10 means they are good for the economy. This is the basis for my second main dependent variable that measures *Immigration Opinion*, where a high score indicates support for immigrants and a low score indicates opposition to immigration. While the ESS also includes items that ask about different aspects of immigration, I use this question since my main argument is based on an economic logic, and thus, effects should be strongest when people are asked how they think immigration impacts the economy. Other aspects

of immigration, such as cultural diversity, might be unaffected by AI exposure in one’s occupation. Someone who is negatively affected by AI might still think that immigrants enrich a country’s cultural life, while they also think that they are bad for the economy. Thus, it is beneficial to separate these aspects and just look at the economic impact of immigration.

Hypothesis 3a and 3b require information on who the respondents would vote for. The ESS asks two questions that aim at this. First, they ask who the respondent voted for in the last election. Additionally, they ask which party the respondent feels closest to. I use the second of these questions for several reasons. Since the first question pertains to the last national election, it could be the case that a respondent has changed their views and would vote for a different party in the next election, especially if the ESS survey is held shortly before an election. This means that the second question is closer to the respondent’s current political affiliation. Consider also that if the survey is held shortly after an election, the first question should yield the same result as the second question (if the respondent has not very quickly changed their mind). And the correlation between the two questions is very high (Pearson’s $r = 0.85$), which means results would not change significantly. In a recent paper, Borwein et al. (2025) combine data from the Chapel Hill Expert Survey and the PopuList database to denote parties that are mainstream left or right, and populist left or right. I use their coding of parties to create dummy variables that denote whether someone voted for a populist right, a populist left, a mainstream right, or a mainstream left party. While their data does not span all countries of the ESS, it still includes 15 countries from across Europe. An overview is given in Appendix 1, Table A1.

3.3 Control Variables and Matching Covariates

I incorporate several theoretically motivated control variables that serve as substantive covariates for regression modeling and balancing variables for the matching. These variables capture fundamental sociodemographic characteristics and political orientations that previous research has established as key determinants of redistributive and immigration preferences and, consequently, vote choice. This enables robust identification of automation-specific effects while ensuring adequate covariate balance across treatment conditions. To allow for a doubly robust method, I use a first set of variables in the matching process and then use a second set of control variables in the regression specifications. This also prevents the overfitting of the regression models to the matched data. Table 1 presents an overview of descriptive statistics for all key variables used in this analysis. These variables are also taken from the ESS, the exact question wording and coding can be found Table A2, in Appendix 1.

The variables discussed in this section are used as covariates in the matching process. *Parental Education* represents a measure of family socioeconomic background and intergenerational transmission of political values (Persson, 2015). Beyond direct material advantages, higher parental education levels typically correlate with greater political sophistication and more liberal social attitudes (Stubager, 2008). Potentially moderating responses to automation-induced labor market disruption, as parental education also affects someone’s education level. Respondents were asked about their mother’s and father’s education, also specified according to ISCED levels. I add the two variables together to get a composite measure of parental education, where higher scores mean parents received higher education. *Income Decile* captures current material position and economic vulnerability, with lower-income individuals theoretically more supportive of redistributive policies due to direct material benefits (Meltzer and Richard, 1981) and potentially more susceptible to immigration-related economic threat perceptions (Scheve and Slaughter, 1999). *Age* captures life-cycle effects and generational differences in political socialization, with older cohorts typically exhibiting more conservative attitudes

toward both economic redistribution and cultural change (Tilley and Evans, 2014). *Gender* represents a well-established predictor of political attitudes, with women demonstrating greater support for welfare state expansion and more positive immigration attitudes in most European societies (Inglehart and Norris, 2000). A person’s *Immigration Background* captures whether a person themselves, or one of their parents, was born in a country that is not the country where the survey is conducted. Individuals with personal or family migration experience typically hold more positive views toward immigration policies and immigrant integration. Failure to control for migration background could particularly bias estimates if automation vulnerability correlates with immigrant concentration in specific occupations or regions. (Maxwell, 2019)

Table 1: Descriptive Statistics of Key Variables

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Median	Pctl(75)	Max
AI Exposure (%)	217,964	54.6	27.6	1	29	67	78	100
Education Level (ISCED)	217,964	4.0	1.8	1	3	4	6	7
AI Score	217,964	3.0	1.0	1	2	3	4	4
AI Score (Binary: Losers)	120,028	0.7	0.4	0	0	1	1	1
Redistribution Preferences	215,746	3.9	1.0	1	3	4	5	5
Immigration Attitudes	211,995	5.1	2.5	0	4	5	7	10
Populist Left Vote (Binary)	46,879	0.04	0.2	0	0	0	0	1
Populist Right Vote (Binary)	46,879	0.1	0.3	0	0	0	0	1
Immigration Background (Binary)	217,964	0.9	0.3	0	1	1	1	1
Gender (Binary: Female)	217,964	0.5	0.5	0	0	1	1	1
Age	217,964	50.4	18.1	14	36	51	65	114
Parents’ Education	217,964	2.8	1.6	1.0	1.0	2.5	4.0	7.0
Income Decile	217,964	5.6	2.8	1	3	6	8	10
Left-Right Scale*	190,006	4.9	2.1	0	4	5	6	9
Political Interest*	217,601	2.4	0.9	1	2	2	3	4
Union Membership*	215,908	0.4	0.5	0	0	0	1	1
Religiosity*	216,840	4.6	3.1	0	2	5	7	10
Year*	216,738	2015.8	4.8	2008	2012	2016	2019	2024
Population Density (NUTS2)*	115,061	3.3	8.3	0.03	0.5	1.0	2.2	77.7
GDP (NUTS2)*	101,892	63.3	81.1	1.0	26.4	39.2	70.5	787.4
Population (NUTS2)*	84,898	1.9	1.7	0.03	1.1	1.4	2.0	12.4

Note: Variables with * only used as control variables in regression models.
Population Density in 100 persons per square kilometer.
GDP in billions of Euro. Population in millions of people.

The variables discussed in this section are used as control variables in the regression estimations. The ideological *Left-Right Scale* serves as a fundamental control for general political predispositions, given its strong association with both redistributive preferences and immigration attitudes across European contexts (Alesina and Giuliano, 2011). *Union Membership* provides a crucial indicator of collective labor organization and class consciousness, historically associated with stronger redistributive preferences and more nuanced responses to immigration based on labor market competition dynamics (Dancygier and Walter, 2015). This variable proves particularly important for automation research, as unionized workers may experience different exposure patterns and possess greater collective resources for responding to technological disruption. *Religiosity* serves as an important control for traditional value orientations and social conservatism. Religious individuals typically exhibit more restrictive immigration attitudes due to concerns about cultural cohesion and traditional values (Scheepers et al., 2002), while also showing distinct patterns of support for redistributive policies that emphasize deservingness criteria (Kahl, 2005). Given that automation vulnerability may interact with traditional value systems in shaping political responses, controlling for religiosity ensures that

estimated effects reflect economic rather than purely cultural mechanisms. *Political Interest* captures variations in political engagement and sophistication that may moderate responses to economic threats. Citizens with higher political interest typically demonstrate more stable and informed political preferences (Dalton, 2019), potentially reducing susceptibility to automation-induced attitude changes. Moreover, politically engaged individuals may be more likely to correctly attribute economic challenges to technological rather than cultural or different economic factors (Hobolt, 2016). I incorporate three regional-level variables measured at the NUTS2 level to capture subnational variation in economic conditions and development patterns that may influence both automation exposure and political attitudes ². *Population Density* accounts for urban-rural differences in economic structure, social networks, and political culture, with higher density areas typically exhibiting more liberal attitudes toward both redistribution and immigration (Scala and Johnson, 2017). *GDP* captures regional economic prosperity and labor market conditions, controlling for the possibility that automation exposure correlates with regional economic performance in ways that independently influence political preferences (Rodríguez-Pose, 2018). *Total Population* controls for regional scale effects and administrative capacity that may influence both technological adoption patterns and policy responsiveness (Peterson, 1981).

In my regression models, I use random and fixed year and country effects, depending on the model, to address two fundamental sources of analytical confounding. Temporal effects control for period-specific shocks affecting political attitudes across all individuals, including economic crises, major political events, or technological developments that might correlate with automation exposure timing. This is particularly relevant in this case as the salience of AI-induced labor market changes increased dramatically with the release of OpenAI’s LLM-based chatbot. This is evidenced by the ever-increasing interest in the terms ‘AI’ and ‘ChatGPT’ as documented by Google Trends data (Google, 2025). Country effects account for persistent cross-national differences in institutional arrangements, political cultures, and baseline automation penetration that could otherwise bias treatment effect estimates (Potrafke, 2009).

These control variables serve as essential balancing covariates in the matching procedures detailed below, ensuring that treatment and control groups exhibit comparable distributions across relevant dimensions. By achieving covariate balance on these fundamental predictors of political attitudes and behavior, the matching and regression estimation strategy isolates automation-specific effects while maintaining the analytical power necessary for robust inference. This dual theoretical and methodological integration strengthens both the internal validity of causal identification and the external validity of findings across diverse European political contexts. For my analysis, I rely on data that ranges from 2015 to 2024 and is from 30 countries that were included in the ESS ³. Those are Austria, Belgium, Switzerland, Czech Republic, Germany, Denmark, Spain, Finland, France, United Kingdom, Greece, Hungary, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Sweden, Slovenia, Estonia, Iceland, Slovakia, Bulgaria, Cyprus, Croatia, Latvia, Lithuania, Montenegro, and Serbia.

²These statistics are taken from the Eurostat website

³Since I only have the party data for 15 countries, the analysis of the voting effects is restricted to those. The countries are: Austria, Belgium, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Netherlands, Norway, Portugal, Spain, Sweden, and the UK

3.4 Model Specification

3.4.1 Initial Specification

The analysis begins with the estimation of baseline effects using linear mixed effects regression on the unmatched dataset. This initial specification provides preliminary evidence of automation’s political impact while accounting for the hierarchical structure of the data, where individuals are nested within countries and survey years. The mixed effects approach allows for random variation in intercepts across countries and years, providing a more flexible modeling framework than traditional fixed effects approaches. The baseline models are included as a robustness check.

$$Y_{ict} = \alpha + \beta_1 \text{AI Score}_{ict} + \mathbf{X}'_{ict}\boldsymbol{\gamma} + u_c + v_t + \epsilon_{ict} \quad (1)$$

where:

α denotes the intercept

Y_{ict} represents the political attitude outcome for individual i in country c at time t

AI Score_{ict} denotes the continuous automation risk measure

$\mathbf{X}_{ict}$ constitutes the vector of individual-level control variables

$u_c \sim N(0, \sigma_u^2)$ denotes random country intercepts

$v_t \sim N(0, \sigma_v^2)$ represents random year intercepts

$\epsilon_{ict} \sim N(0, \sigma^2)$ denotes the individual-level error term

The random effects structure is specified as:

$$u_c \sim N(0, \sigma_u^2) \text{ and } v_t \sim N(0, \sigma_v^2) \quad (2)$$

with the assumption that u_c , v_t , and ϵ_{ict} are mutually independent.

The coefficient β_1 provides an initial estimate of automation’s average association with political preferences, treating the AI Score (the 4-point scale ranging from 1-4) as a continuous measure rather than categorical classifications. The vector $\mathbf{X}_{ict}$ includes parental education, left-right political orientation, gender, age, income decile, and union membership. The random intercepts approach allows for country-specific and year-specific baseline levels while assuming that the treatment effect (β_1) is constant across these groupings. This specification is particularly appropriate given the cross-national, cross-temporal nature of the ESS data, as it accounts for unmeasured country-level factors (such as institutional differences, political cultures, or economic structures) and period-specific shocks (such as economic crises or major events such as the release of ChatGPT in November 2022) without requiring the estimation of numerous fixed effect parameters.

However, potential confounding from unobserved heterogeneity across automation risk categories necessitates the subsequent matching analysis for robust causal identification. The mixed effects specification provides valuable preliminary evidence while serving as a foundation for the more rigorous causal identification strategy that follows (Stuart, 2010).

3.4.2 Matching

The matching strategy operationalizes automation risk through a binary treatment indicator that isolates the contrast in labor market vulnerability. Following the theoretical emphasis on differential automation impacts across skill-automation combinations, the treatment assignment leverages the extreme categories of the four-point classification:

$$T_{ict} = \begin{cases} 1 & \text{if AI_Score}_{ict} = 4 \text{ (Low Skill, High AI Exposure)} \\ 0 & \text{if AI_Score}_{ict} = 1 \text{ (High Skill, High AI Exposure)} \\ \text{Missing} & \text{otherwise} \end{cases} \quad (3)$$

This specification compares individuals facing maximal automation threat (treatment group) with those positioned to benefit from AI complementarity (control group), thereby isolating the effect of automation-induced labor market vulnerability. The binary treatment framework enables the implementation of matching methods while preserving theoretical coherence regarding automation’s heterogeneous labor market impacts. The selection of the final matching specification proceeded through a systematic evaluation of multiple methodological approaches to ensure optimal covariate balance while maintaining adequate statistical power, as is standard in the literature (Ho et al., 2007). This comprehensive specification testing encompassed variations across four critical dimensions: distance measures, caliper restrictions, matching ratios, and treatment definitions.

I compared logistic regression propensity scores against alternative distance metrics, including Generalized Boosted Models (GBM), Random Forest, Neural Networks, and Mahalanobis distance. While machine learning approaches (GBM, Random Forest) demonstrated marginally superior predictive performance in propensity score estimation, they yielded comparable covariate balance outcomes when combined with appropriate caliper restrictions. The logistic regression specification was retained for its interpretability and computational efficiency while achieving equivalent balance performance. Additionally, I systematically evaluated caliper restrictions. This revealed a clear trade-off between sample retention and covariate balance quality. Specifications ranged from no caliper restriction (31,085 matched pairs, SMD = 1.64) through standard restrictions (0.2 caliper: 27,031 pairs, SMD = 0.172) to increasingly stringent constraints. The ultra-tight caliper specification (0.05) achieved very good balance (SMD = 0.047) while retaining substantial analytical power (25,871 matched pairs), representing the optimal balance between statistical efficiency and inferential capacity. An evaluation of 1:1, 1:2, and 1:3 matching ratios demonstrated that increased control matching marginally improved statistical precision without substantively altering effect estimates. The 1:1 specification was maintained to preserve the clear counterfactual interpretation central to the theoretical framework while avoiding potential complications from differential weighting schemes. Lastly, alternative treatment specifications comparing extreme categories (AI Score 1 vs. 4) against broader categorizations (AI Scores 1,2 vs. 3,4) yielded substantively consistent results, validating the theoretical motivation for focusing on maximal automation risk contrasts. The extreme category specification was retained for its theoretical clarity and stronger causal identification properties. Table A3 in Appendix 2 discloses the different matching specifications and their results in more detail.

The final specification achieved standardized mean differences below 0.05 across all covariates while maintaining sufficient sample size for robust statistical inference, and is detailed in the following. The propensity score represents the conditional probability of treatment assignment given observed covariates and incorpo-

rates exact matching constraints on the country level to achieve covariate balance between treatment and control groups:

$$e(\mathbf{X}_{ict}, c, t) = P(T_{ict} = 1 | \mathbf{X}_{ict}, \text{Country}_c) = \text{logit}^{-1}(\mathbf{X}'_{ict}\boldsymbol{\beta}) \quad (4)$$

where $\mathbf{X}_{ict}$ includes all matching covariates (parental education, immigration background, age, gender, income decile, union membership) and $\boldsymbol{\beta}$ represents the logistic regression coefficients. The estimation incorporates exact matching constraints on country indicators to ensure cross-national comparability, effectively estimating propensity scores within country strata.

The final specification uses nearest-neighbor matching with replacement using an ultra-tight caliper restriction ($c = 0.05$) on propensity score differences. This specification prioritizes covariate balance over sample retention, employing the following matching protocol:

$$\text{Match}(i, j) = \arg \min_{j \in \{T_j=0\}} |e(X_{ict}) - e(X_{jct})| \quad (5)$$

subject to:

$$|e(X_{ict}) - e(X_{jct})| \leq 0.05 \quad (6)$$

$$\text{Country}_i = \text{Country}_j \quad (7)$$

where $e(X_{ict})$ represents the estimated propensity score for individual i in country c at time t , and the constraints ensure that matches occur only within the same country strata and within the specified caliper distance. The ultra-tight caliper restriction achieves standardized mean differences below 0.05 across all covariates while retaining sufficient sample size for robust statistical inference. An exact balance table is presented in Table A4 in Appendix 2.

Figure 3 presents the distribution of estimated propensity scores across treatment and control groups in both pre-matching and post-matching configurations. The pre-matching panel reveals substantial distributional differences between automation-vulnerable workers (treatment group) and automation-complemented workers (control group), with limited overlap in the propensity score support region. This distributional divergence indicates significant baseline differences in observed covariates that could bias naive treatment effect estimates. The post-matching panel demonstrates successful achievement of common support through the ultra-tight caliper restriction, with near-identical propensity score distributions across treatment conditions. The improvement in distributional overlap validates the matching procedure's effectiveness in creating comparable treatment and control groups, thereby satisfying the fundamental assumption of ignorable treatment assignment conditional on observed covariates.

Figure 4 provides a comprehensive visualization of standardized mean differences across all matching covariates before and after the matching procedure. The love plot format enables systematic evaluation of balance improvement, with the vertical reference line at zero representing perfect covariate balance and the conventional threshold line at 0.1 indicating acceptable balance for causal inference. Pre-matching standardized mean differences (red points) demonstrate substantial imbalances across multiple dimensions. These imbalances reflect the systematic differences in automation vulnerability patterns across parental educational levels, geographic contexts, and other demographic variables that motivate the matching approach. Post-matching standardized mean differences (blue points) cluster tightly around zero, with all covariates achieving

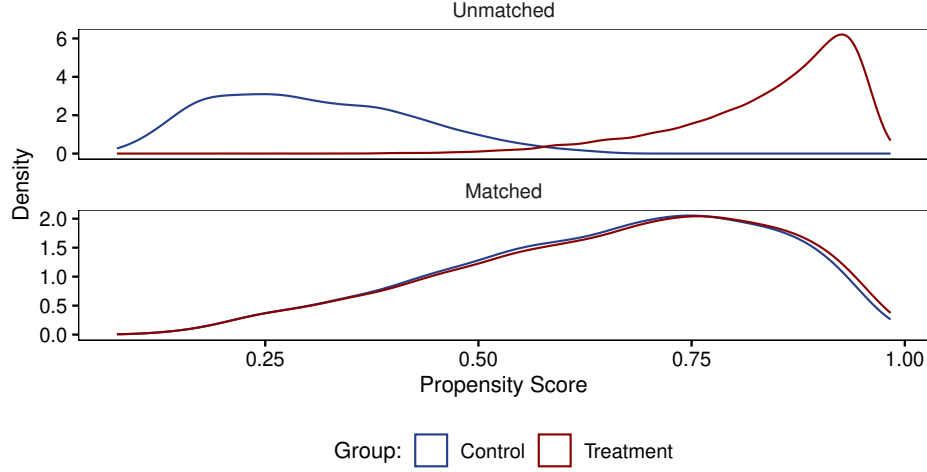


Figure 3: Density of Propensity Scores in Treatment and Control Group Before and After Matching

standardized mean differences below 0.05. This balance performance provides strong evidence for the successful elimination of selection bias on observed confounders, thereby strengthening the causal interpretation of subsequent treatment effect estimates. The consistent balance improvement across diverse covariate types (continuous, binary, and categorical indicators) demonstrates the robustness of the matching specification to different variable structures and scaling properties.

3.4.3 Analysis on Matched Dataset

Analysis of the matched dataset employs weighted least squares regression that incorporates propensity score matching weights to ensure appropriate statistical inference while maintaining the causal identification framework. The matched sample specification uses year fixed effects:

$$Y_{ict}^{\text{matched}} = \alpha + \tau \cdot T_{ict} + \mathbf{X}_{ict}'\gamma + \lambda_t + \epsilon_{ict} \quad (8)$$

where the parameters are estimated by minimizing the weighted sum of squared residuals:

$$\min_{\alpha, \tau, \gamma, \lambda} \sum_{i, c, t} w_{ict} (Y_{ict}^{\text{matched}} - \alpha - \tau \cdot T_{ict} - \mathbf{X}_{ict}'\gamma - \lambda_t)^2 \quad (9)$$

where Y_{ict}^{matched} represents the political attitude outcomes for individual i in country c at time t within the matched sample, and τ denotes the average treatment effect (ATE). The binary treatment indicator T_{ict} distinguishes between automation-vulnerable workers (low-skilled, high AI exposure; $T_{ict} = 1$) and automation-complemented workers (high-skilled, high AI exposure; $T_{ict} = 0$). The covariate vector $\mathbf{X}_{ict}'$ includes left-right orientation, political interest, religiosity, NUTS2 population density, NUTS2 GDP, and NUTS2 population to control for any remaining imbalance after matching. Year fixed effects (λ_t) control for temporal shocks. Country fixed effects are not included due to exact matching on country strata, which ensures within-country comparisons. Propensity score weights (w_{ict}) derived from the matching procedure ensure that statistical inference appropriately accounts for the matching process while preserving the matched

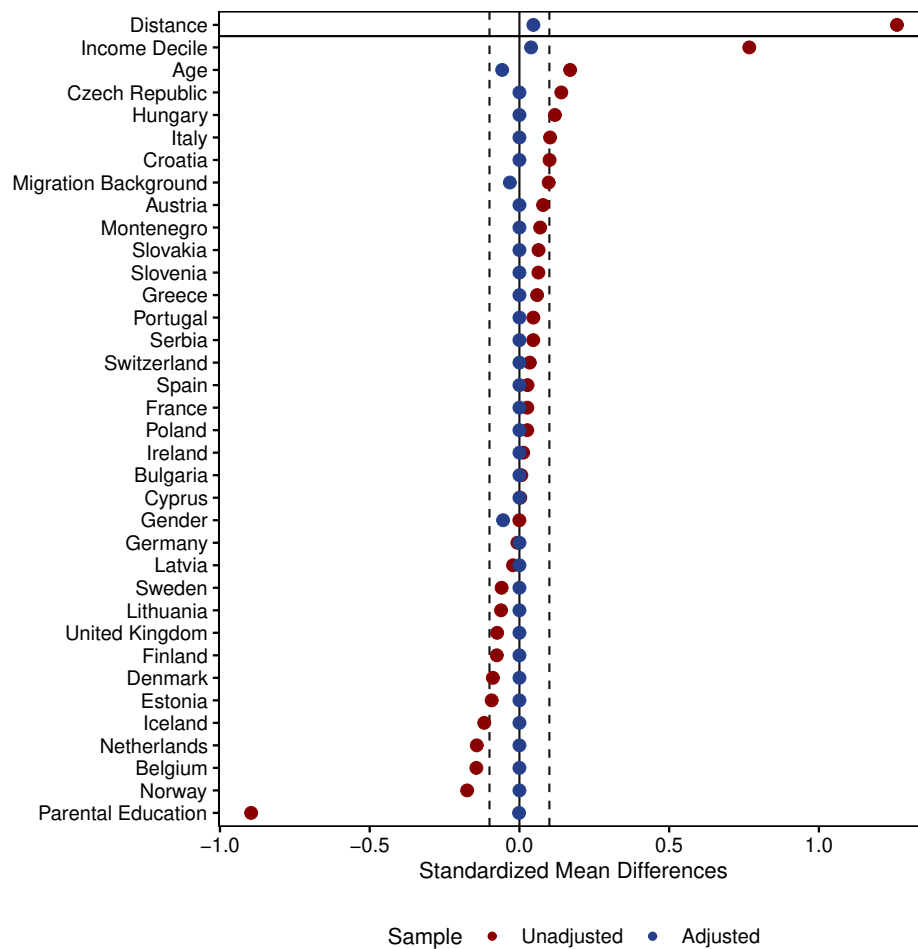


Figure 4: Love Plot of Pre- and Post-Matching Balance

sample's covariate balance properties. These weights correct for the sampling design induced by the matching algorithm and maintain the validity of standard error calculations. Robust standard errors are used to accommodate the potential heteroskedasticity arising from the propensity score weighting scheme.

Additionally, as a robustness check, I calculate average treatment effects using marginal effects methods that account for matching weights and the complete covariate structure (Schafer and Kang, 2008). The ATE estimator is based on a fully interacted model:

$$Y_{ict} = \alpha + \tau \cdot T_{ict} + \mathbf{X}'_{ict}\boldsymbol{\gamma} + T_{ict} \times \mathbf{X}'_{ict}\boldsymbol{\delta} + \beta \cdot \text{year}_t + \epsilon_{ict} \quad (10)$$

estimated with propensity score weights w_{ict} , where the interaction terms allow for heterogeneous treatment effects across covariates. The ATE is then calculated as:

$$\hat{\tau}^{\text{ATE}} = \frac{1}{N_1} \sum_{i:T_{ict}=1} w_{ict} \left[Y_{ict} - \hat{Y}_{ict}^0(\mathbf{X}_{ict}, \text{year}_t) \right] \quad (11)$$

where N_1 represents the number of treated units, w_{ict} denotes matching weights, and $\hat{Y}_{ict}^0(\mathbf{X}_{ict}, \text{year}_t)$ constitutes the predicted counterfactual outcome for treated individuals conditional on their observed covariates and year, with the treatment indicator set to zero. This approach leverages the interaction terms to estimate heterogeneous treatment effects and then averages these effects across the treated population, accounting for both the matching weights and the full covariate structure.

3.4.4 Mediation Analysis

To test the theoretical pathways linking automation vulnerability to populist voting behavior (Hypotheses 3a and 3b), I employ causal mediation analysis following the framework developed by Imai et al. (2010b). This approach allows for the decomposition of the total treatment effect into direct effects and indirect effects operating through specific mediating variables, providing insights into the mechanisms underlying automation's political consequences.

The mediation analysis estimates three key quantities for each pathway: the Average Causal Mediation Effect (ACME), which captures the indirect effect of automation vulnerability operating through the proposed mediator; the Average Direct Effect (ADE), which represents the remaining direct effect after accounting for the mediating pathway; and the Total Effect, which equals the sum of ACME and ADE. The proportion mediated is calculated as ACME divided by the Total Effect, indicating the percentage of the total effect that operates through the specific mediating mechanism.

The analysis follows a two-stage estimation procedure. First, I estimate mediator models that regress each proposed mediating variable (redistribution preferences and immigration attitudes) on the treatment indicator and the full set of control variables:

$$M_{ict} = \alpha_M + \tau_M T_{ict} + \mathbf{X}'_{ict}\boldsymbol{\gamma}_M + \delta_{Mc} + \lambda_{Mt} + \epsilon_{Mict} \quad (12)$$

Second, I estimate outcome models that regress the voting outcomes on both the treatment and the mediating variables:

$$Y_{ict} = \alpha_Y + \tau_Y T_{ict} + \beta M_{ict} + \mathbf{X}'_{ict}\boldsymbol{\gamma}_Y + \delta_{Yc} + \lambda_{Yt} + \epsilon_{Yict} \quad (13)$$

where M_{ict} represents the mediating variable, T_{ict} denotes the binary treatment indicator, and both

models include country and year fixed effects along with the full control variable structure also used in the models for H1 and H2.

I test four distinct mediation pathways: (1) redistribution preferences mediating the effect on populist left voting, (2) immigration attitudes mediating the effect on populist right voting, and (3,4) two cross-pathways examining whether each mediator influences the alternative voting outcome. This comprehensive approach provides insights into both theoretically expected pathways and potential cross-cutting effects.

The mediation effects are estimated using quasi-Bayesian simulation methods with 1,000 iterations to construct confidence intervals that appropriately account for uncertainty in both stages of the estimation process (Imai et al., 2010a). To address concerns about unmeasured confounding that could bias mediation estimates, I conduct sensitivity analyses using the approach developed by Imai et al. (2010b). These analyses assess how robust the mediation findings are to violations of the sequential ignorability assumption by estimating sensitivity parameters that would be required to alter the substantive conclusions. The results of this are presented in Table A5 in Appendix 2.

The mediation analysis is conducted on the matched dataset to ensure that the treatment-mediator and treatment-outcome relationships are identified through the same causal identification strategy established in the matching procedure. This approach maintains consistency across all models while leveraging the improved covariate balance achieved through matching to strengthen the plausibility of the sequential ignorability assumption underlying causal mediation analysis (Imai et al., 2011).

4 Results

Before moving to the inferential analysis to test my hypotheses, I take a descriptive look at the data to better understand it and gain additional insight and context.

4.1 Descriptive Analysis

The analysis of AI exposure across education levels provides important validation for the study’s theoretical framework. A key concern for this research is whether education simply serves as a proxy for AI exposure, which would undermine the distinction between low-skilled and high-skilled workers in highly exposed occupations.

Figure 5 demonstrates that education is not a proxy for AI exposure, as AI exposure remains relatively constant across all education levels. The thick black lines showing mean exposure scores are essentially at the same level across ISCED categories. This finding is crucial because it supports the theoretical argument that AI affects both high-skilled and low-skilled occupations. This is a departure from previous research that has explored the impact of AI on occupations and has come to the conclusion that jobs that require highly skilled employees and workers are those most highly exposed (Brynjolfsson et al., 2018; Webb, 2020).

The distribution patterns reveal additional important characteristics. The vertical lines extend to the minimum and maximum values and show that every education level contains respondents with both the highest and lowest possible AI exposure scores. This wide range within each education category confirms that technological exposure cannot be predicted by educational attainment alone.

Some variation does appear in the interquartile ranges across education levels. For example, the 25th percentiles for ISCED levels 4 and 5 (lower and upper tier upper secondary education) are slightly lower than other categories, while the 7th ISCED level (higher tertiary education) shows a somewhat tighter distribution

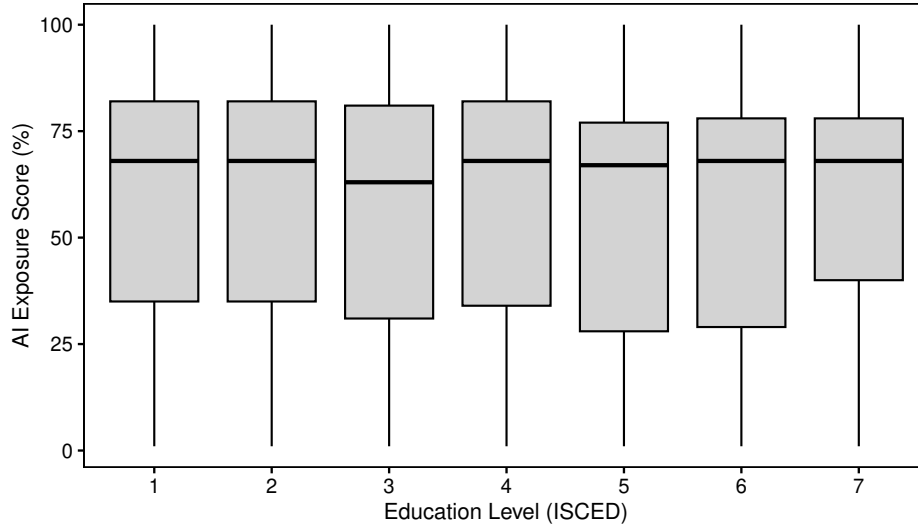


Figure 5: Distribution of AI Exposure Score across ISCED Education Levels

around the mean. However, these differences are relatively minor compared to the overall consistency in mean exposure levels.

This pattern validates the approach of creating an AI Score that combines exposure and education, as it shows that it is not just highly-skilled respondents who are predominantly exposed to AI. The finding that AI exposure cuts across educational boundaries supports the theoretical framework distinguishing between workers who face automation threats versus those positioned to benefit from AI complementarity.

To add to the descriptive analysis of the data, I also explore how occupational AI exposure varies between European countries. Figure 6 reveals substantial differences in AI exposure patterns across European countries, highlighting the importance of controlling for country-specific factors in the analysis. Mean AI exposure varies by over 10 points, from 52.58 in Iceland to 63.66 in Montenegro. This range demonstrates that AI exposure might reflect systematic differences in economic structure, industrial composition, and occupational patterns.

Several geographic patterns emerge from the data. Eastern European countries (Montenegro, Serbia, Croatia) and some Southern European countries (Italy, Greece) tend to show higher exposure scores, while Nordic countries (Iceland, Finland) and Baltic countries (especially Lithuania) display lower exposure levels. These patterns likely reflect differences in industrial structure and economic development. The necessity of including country fixed (or random) effects and matching exactly on country indicators becomes clear from this variation. Without controlling for country-specific factors, estimated effects could be confounded by systematic differences in national economic structures, institutions, or development levels that correlate with both AI exposure and political attitudes.

The bottom panel of Figure 6 shows the share of workers classified as vulnerable to AI displacement (those receiving an AI score of 4). The cross-national variation in vulnerability is even more pronounced, ranging from 31.40% in Iceland to 58.05% in Montenegro, a difference of 26.65 percentage points. This larger range might occur because vulnerability depends on the interaction between AI exposure and education levels within each country. This already indicates that the same countries that had a high exposure score also have the highest share of vulnerable people. This is not only confirmed by examining the colors of the two maps,

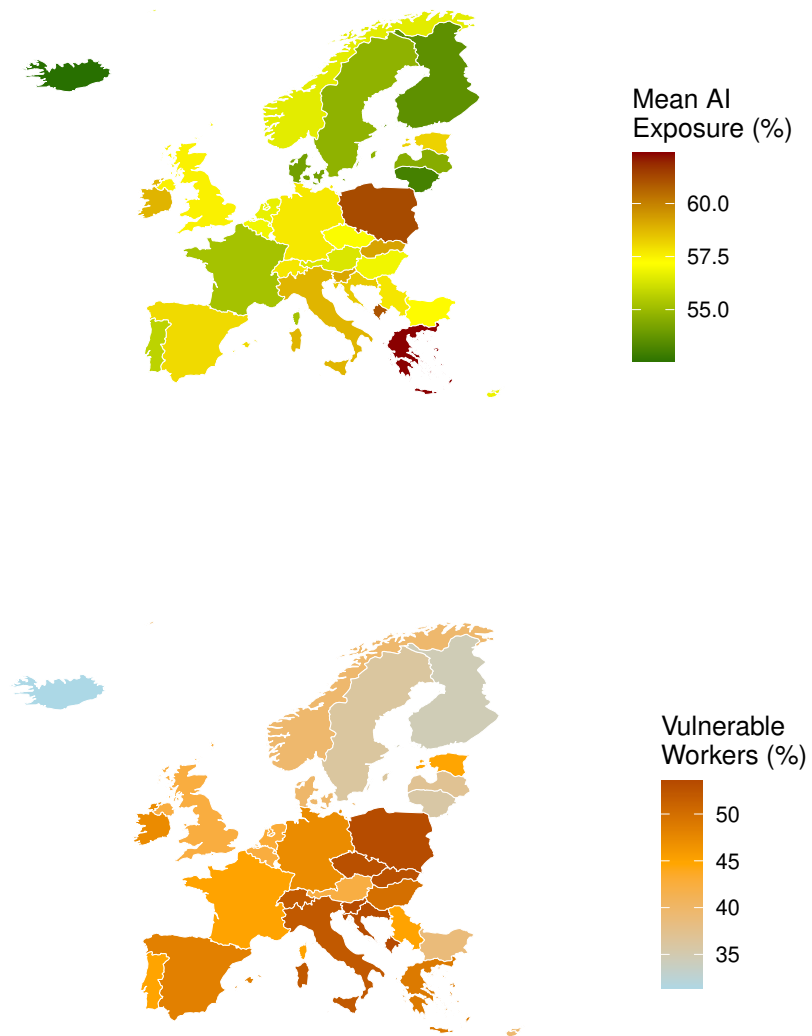


Figure 6: Map of Mean AI Exposure and AI Vulnerability

but also by the fact that the Pearson correlation between the countries’ exposure scores and vulnerability shares is 0.8, which is very high.

These cross-national patterns provide an important context for interpreting the individual-level results. Countries with higher concentrations of vulnerable workers may experience different political dynamics around automation and related policy issues. The systematic variation also underscores why the analysis focuses on within-country comparisons rather than relying on cross-national variation for identification.

4.2 Inferential Analysis

Table 2 presents the initial OLS regression results using the full, unmatched dataset to establish preliminary evidence of automation’s association with political attitudes. I use the AI Score variable, where, to reiterate, a score of 4 indicates a highly negative effect, a score of 3 a muted negative effect, a score of 2 a muted positive effect, and a score of 1 a highly positive effect of automation.

It is worth noting that in interpreting the regression results, I focus exclusively on the coefficients of the main explanatory variable, rather than the control variables. This approach follows established methodological principles in political science research that caution against interpreting control variables, as their coefficients represent partial associations after adjusting for other variables in the model, not causal effects (Cinelli and Hazlett, 2020; Westfall and Yarkoni, 2016). These variables are included to adjust for potential confounding factors that might influence both the dependent and independent variables, but their specific effects are not the focus of the theoretical framework being tested. This holds for all my models, also after matching.

Model (1) results in a statistically significant relationship with the dependent variable measuring redistribution preferences. The positive association indicates that a higher score on the AI score variable (and thus being displaced by AI) results in more support for redistribution preferences. Model (2) reveals a negative association with immigration attitudes. Thus, people who score high on the AI score variable (i.e. those proposed to be predominantly negatively affected by AI) have a more unfavorable opinion about the economic effects of immigrants. These initial findings suggest that higher AI score categories are associated with higher support for redistributive policies and less positive views toward immigration’s economic impact, and vice versa. This is in line with hypotheses H1 and H2. However, these preliminary estimates should be interpreted with considerable caution, as they represent associations rather than causal effects. For robustness, I also calculate models using fixed effects for country and year, with country-clustered standard errors, but results remain the same. They are presented in Table A6, in Appendix 3.

Table 3 presents the treatment effect estimates using the matched dataset, which provides substantially more reliable causal inference due to the achieved covariate balance. The results reveal significant and substantively meaningful effects of automation vulnerability on both political attitudes. Workers and employees in the treatment group (low-skilled, high AI exposure) demonstrate significantly higher support for redistributive policies compared to the control group (high-skilled, high AI exposure). The positive coefficient indicates that automation-vulnerable workers increase their support for government redistribution. Conversely, the same treatment group exhibits significantly more negative attitudes toward immigration’s economic impact. As a robustness check, I also calculate the models using the linear mixed effects approach, with year and country random effects, but results do not change significantly. These models are detailed in Table A7, in Appendix 3.

Figure 7 provides an intuitive visualization of the substantive magnitude of these treatment effects by comparing predicted outcomes for the treatment and control conditions while holding all other variables at their

Table 2: Linear Mixed Effects Models (Unmatched Sample, Random Effects)

	<i>Dependent variable:</i>	
	Redistribution Preferences	Immigration Attitudes
	(1)	(2)
AI Automation Risk	0.038*** (0.002)	−0.247*** (0.005)
Parental Education	−0.036*** (0.002)	0.148*** (0.004)
Left-Right Scale	−0.100*** (0.001)	−0.092*** (0.003)
Gender (Female)	0.099*** (0.004)	−0.178*** (0.011)
Age	0.001*** (0.0001)	0.001*** (0.0003)
Income Decile	0.039*** (0.001)	−0.079*** (0.002)
Union Member	0.111*** (0.005)	0.027** (0.012)
Country Random Effects	Yes	Yes
Year Random Effects	Yes	Yes
Model	Linear Mixed	Linear Mixed
Observations	182,848	182,848
Log Likelihood	−248,722.400	−406,336.300
Akaike Inf. Crit.	497,466.700	812,694.700
Bayesian Inf. Crit.	497,578.000	812,806.000

Note:

*p<0.1; **p<0.05; ***p<0.01

Standard errors in parentheses.

Random intercepts for country and year.

Table 3: Treatment Effects of Automation Risk on Political Attitudes (Matched Sample)

	<i>Dependent variable:</i>	
	Redistribution Preferences	Immigration Attitudes
	(1)	(2)
Treatment	0.102*** (0.017)	−0.531*** (0.038)
Left-Right Scale	−0.141*** (0.004)	−0.104*** (0.010)
Political Interest	0.010 (0.010)	0.507*** (0.023)
Union Member	0.100*** (0.017)	−0.070* (0.038)
Religiosity	0.019*** (0.003)	−0.001 (0.006)
Populalation Density (NUTS2)	0.00001 (0.00001)	−0.00004** (0.00002)
GDP (NUTS2)	−0.00000*** (0.00000)	0.00000*** (0.00000)
Population (NUTS2)	0.00000*** (0.000)	−0.00000*** (0.00000)
Constant	4.276*** (0.059)	5.030*** (0.128)
Year FE	Yes	Yes
Weights	PSM	PSM
Observations	14,474	14,384
R ²	0.091	0.094
Adjusted R ²	0.090	0.093

Note:

*p<0.1; **p<0.05; ***p<0.01

Robust standard errors (HC3) in parentheses.

Models estimated using matched data with propensity score weights.

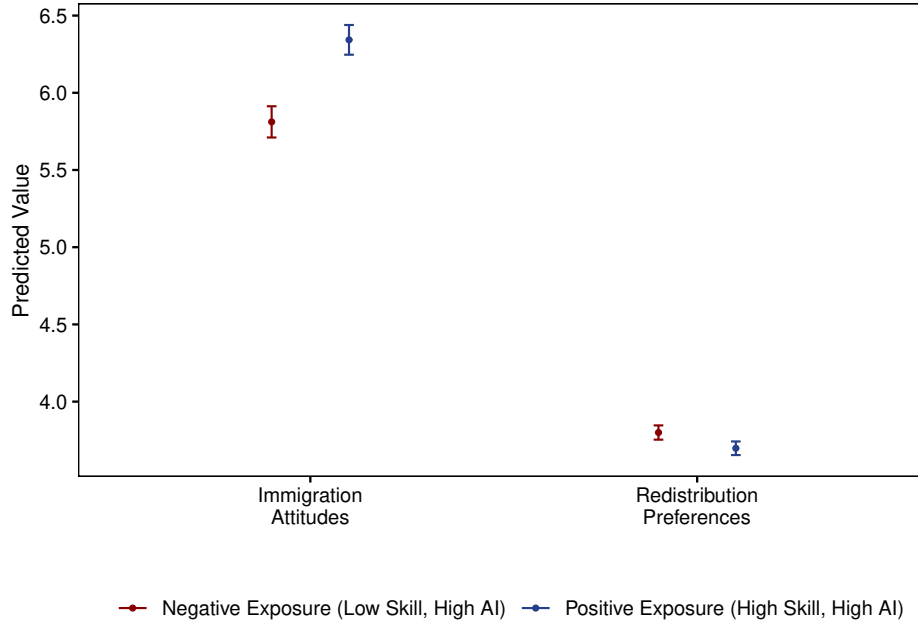


Figure 7: Comparison of Scenarios using OLS results based on the matched data

median values. The figure clearly illustrates the divergent political responses between automation-vulnerable workers and those positioned to benefit from AI complementarity. For redistribution preferences, workers facing negative automation exposure (low-skilled, high AI exposure) show meaningfully higher predicted support compared to those experiencing positive exposure (high-skilled, high AI exposure). For immigration attitudes, the visualization reveals an even more pronounced divergence, with automation-vulnerable workers displaying substantially more negative views toward immigration’s economic impact. The separation between the confidence intervals for both outcomes reinforces the robustness of the estimated treatment effects and suggests that automation exposure creates distinct political preference patterns depending on skill level and expected labor market consequences.

Table 4 presents the ATE estimates derived from marginal effects analysis on the matched dataset, providing a complementary perspective to the regression coefficients by estimating the expected change in outcomes if all treated individuals were moved from the treatment to the control condition. The ATE estimates closely align with the regression results while offering a more intuitive interpretation of the treatment’s causal impact.

For redistribution preferences, the ATE of 0.132 indicates that automation vulnerability increases support for government redistribution by approximately 0.106 points on the 5-point scale. Considering the standard deviation of 1 point and that the 25th and 75th percentiles are only 2 points apart, this can make a substantial difference. The ATE for immigration attitudes also reveals a substantive impact, with automation-vulnerable workers showing attitudes that are 0.528 points lower on the 10-point scale compared to their high-skilled counterparts facing similar AI exposure. Again, consider that the standard deviation is just 2.5 and that the 25th and 75th percentiles are just 3 points apart, a 0.5 point difference is a considerable jump (or drop). The narrow confidence intervals for both estimates demonstrate the precision of the treatment effect estimates, while the consistent statistical significance across both outcomes reinforces the robustness of the findings.

Table 4: Average Treatment Effects of Automation Risk (Matched Sample)

	<i>Dependent variable:</i>	
	Redistribution Preferences	Immigration Attitudes
	(1)	(2)
Average Treatment Effect	0.106***	-0.528***
95% CI	[0.072, 0.14]	[-0.603, -0.452]
p-value	< 0.001	< 0.001
N (treated)	25,871	25,871

Note:

*p<0.1; **p<0.05; ***p<0.01

Robust standard errors (HC3) in parentheses.

ATEs estimated using marginal effects at the means.

4.3 Mediation Analysis

The mediation analysis reveals that immigration attitudes, rather than redistribution preferences, serve as the primary pathway linking automation vulnerability to populist voting behavior. Table 5 presents the comprehensive mediation results, while Figure 8 illustrates the hypothesized pathways and their relative strength.

The results provide strong support for Hypothesis 3b, demonstrating that immigration attitudes constitute a substantial mediating pathway between automation vulnerability and populist right voting. The Average Causal Mediation Effect (ACME) of 0.017 is highly significant, indicating that automation-vulnerable workers' more negative immigration attitudes contribute meaningfully to their increased likelihood of supporting populist right parties. Importantly, this mediation effect accounts for 36.6% of the total effect of automation vulnerability on populist right voting, representing more than one-third of the overall relationship. The substantial proportion mediated suggests that immigration attitudes are not merely a correlate of automation vulnerability but rather a key mechanism through which economic insecurity translates into political behavior.

Table 5: Mediation Analysis: Pathways from Automation Vulnerability to Populist Voting

Pathway	Mediation Effects			
	ACME	ADE	Total Effect	Prop. Mediated
Immigration Attitudes → Populist Right	0.017***	0.036***	0.047***	36.6%***
Redistribution Preferences → Populist Left	0.0002***	0.003.	0.003*	7.4%.
Immigration Attitudes → Populist Left	0.003***	0.010	0.013*	23.6%*
Redistribution Preferences → Populist Right	0.001.	0.050***	0.051***	2.0%.

Notes: ACME = Average Causal Mediation Effect (indirect effect through attitudes). ADE = Average Direct Effect (remaining direct effect of automation vulnerability). Total Effect = ACME + ADE. Prop. Mediated = ACME/Total Effect. Significance: *** p<0.001, ** p<0.01, * p<0.05, . p<0.1. Based on 1,000 simulations using quasi-Bayesian confidence intervals. Sample size: 3,158 matched observations.

In contrast, the evidence for Hypothesis 3a is considerably weaker. While redistribution preferences do show a statistically significant mediation effect for populist left voting, the substantive magnitude is minimal, accounting for only 7.4% of the total effect. This small proportion mediated indicates that redistribution preferences play a limited role in connecting automation vulnerability to populist left support, suggesting that other mechanisms beyond redistributive demands drive automation-vulnerable workers toward populist

left parties. The statistical significance is also not as convincing, and the sensitivity analysis in Appendix 2 (Table A5) reveals that the effect is not as robust as the effect of Hypothesis 3b. Thus, one has to conclude that Hypothesis 3a is not supported by these findings.

The analysis also reveals interesting cross-pathway effects that illuminate the complexity of automation’s political consequences. Immigration attitudes demonstrate a significant mediation effect for populist left voting as well, accounting for 23.6% of the total effect. This finding suggests that automation-vulnerable workers’ negative immigration attitudes may contribute to support for populist parties across the political spectrum, not exclusively on the right. Conversely, redistribution preferences show minimal mediation effects for populist right voting (2.0% of total effect), reinforcing the asymmetric nature of these mediating pathways. The sensitivity analysis (Table A5, Appendix 2) again shows that these effects are not as robust as the effect for H3b.

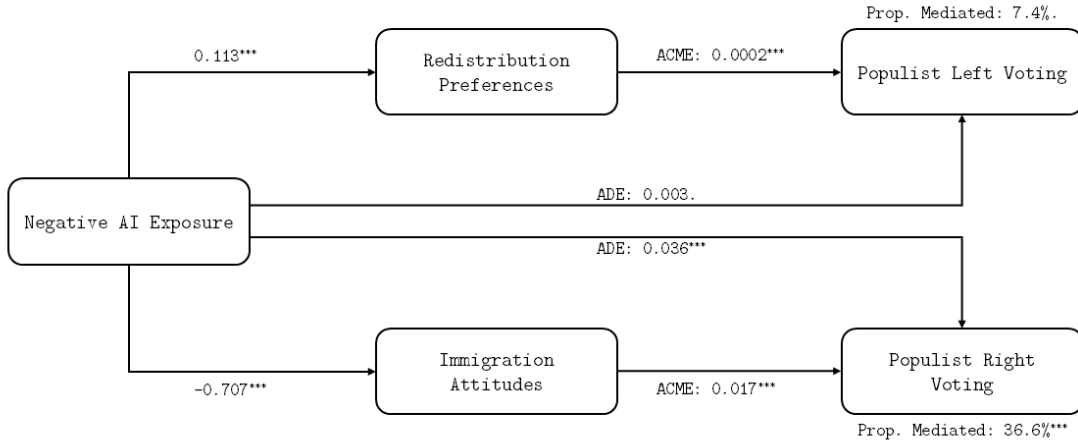


Figure 8: Mediation Pathways from Automation Vulnerability to Populist Voting

Figure 8 visualizes these mediation pathways, clearly illustrating the differential strength of the proposed mechanisms. The diagram demonstrates that automation vulnerability’s primary political consequence operates through immigration attitudes leading to populist right voting, while the redistributive pathway to populist left voting plays a more limited role. The pattern of results suggests that automation-induced economic insecurity is primarily channeled through cultural and symbolic concerns about immigration rather than direct material demands for redistribution. This finding aligns with theoretical arguments about the indirect effects of automation (Gallego and Kurer, 2022), where objective labor market threats are redirected toward more politically salient targets such as immigration. Automation vulnerability appears to activate perceptions of group competition and cultural threat that populist right parties are particularly well-positioned to exploit through their nationalist and nativist platforms. Additionally, the cross-pathway effect on populist left voting suggests that immigration attitudes may serve as a more general expression of anti-establishment sentiment among economically threatened workers. These findings have important impli-

cations for understanding the political consequences of technological change and the strategic positioning of political parties in response to automation-induced labor market disruption, which I discuss in more detail below.

5 Conclusion

After giving a brief summary of the main arguments and findings, I highlight the limitations of this paper before discussing the scientific and practical implications that this analysis reveals.

This paper examined how AI automation influences political attitudes and behavior by examining whether workers who face negative automation exposure have distinct political preferences that translate into specific political behavior. The argument was based on the theoretical expectation that AI automation creates winners and losers depending on the interaction between occupational AI exposure and skill level: high-skilled workers in AI-exposed occupations benefit from complementarity effects (the winners), while low-skilled workers in exposed occupations face displacement (the losers).

Using propensity score matching analysis of ESS data combined with Webb’s (2020) AI exposure measure, the study finds evidence that automation vulnerability significantly shapes political attitudes. Workers and employees classified as AI automation losers show higher support for redistributive policies and more negative attitudes toward immigration compared to automation winners. A follow-up mediation analysis reveals that these attitudinal changes translate into distinct voting patterns, with immigration attitudes serving as the primary pathway linking automation vulnerability to populist right voting (mediating 37% of the effect), while redistribution preferences only play a limited role in explaining populist left voting. Interestingly, anti-immigration attitudes also mediate the effect of AI automation vulnerability on populist left voting, explaining 24%. I go on to discuss the limitations of this paper and the practical implications of the findings, and how these aspects can inform future research.

5.1 Limitations

There are a number of limitations that have to be kept in mind when interpreting these findings. First of all, they should be interpreted as preliminary, given that AI automation remains in its early stages of development and adoption. While AI patent applications have grown exponentially in recent years, the full labor market impacts of artificial intelligence have yet to materialize across most sectors (Brynjolfsson et al., 2021). The period under study (2015-2024) captures the initial phases of AI adoption, meaning the political consequences observed here may intensify, diminish, or evolve in unexpected directions as AI capabilities advance and become more widely integrated into workplace practices⁴.

Additionally, the use of education as a proxy for skill level represents a significant methodological limitation that may obscure important heterogeneity in how workers experience AI automation. Education captures formal qualifications but fails to account for domain-specific expertise, on-the-job training, adaptability to new technologies, or the complementary skills that research suggests are crucial for benefiting from AI integration (Choudhury et al., 2020). Future research would benefit from more granular measures of worker capabilities and their specific relevance to AI-augmented work environments. While my approach offers a

⁴In a similar vein, one could suspect that longstanding grievances drive these results instead of AI automation specifically. To address this, I recalculate the effects for the two years before and after the release of ChatGPT. Effects do not differ significantly. They are presented in Table A8, in Appendix 3.

first idea of how to measure the distributional impact of AI automation, further refinements of this approach are necessary ⁵.

On a more general level, this analysis treats technological change as an independent variable affecting political attitudes, but this represents a considerable simplification of the complex socio-economic processes underlying technology development and adoption, and how the accompanying changes are dealt with. Countries, regions, and firms vary substantially in their capacity and incentives to develop, adopt, and implement AI technologies, with these differences potentially shaping both the pace of automation and its political consequences (Acemoglu and Restrepo, 2019). The assumption of exogenous technological change neglects how political preferences and institutional arrangements may themselves influence technology trajectories.

Moreover, this study identifies significant effects of automation vulnerability on redistribution and immigration preferences, and that, at least immigration attitudes, mediate the effect of negative AI automation on populist voting. Yet, only parts of the effect on voting is explained by these attitudes, and a substantial part remains. This hints at the fact that other attitudes might play an equally important, or even more important, role in explaining the effect of negative AI exposure on populist voting. While this paper thus offers a first idea of how the causal mechanism behind the effect might look like, more research is needed to fully understand this well-documented phenomenon.

5.2 Discussion

Even considering the limitations outlined above, the findings have a number of interesting implications I want to discuss in the following. First, the fact that immigration attitudes mediate a substantially larger proportion of automation’s political effects compared to redistribution preferences may reflect the distinctive institutional context of European welfare states. Unlike the United States, European countries generally maintain more comprehensive social safety nets that may buffer workers against immediate economic insecurity. When basic economic protection is already institutionalized, automation-induced anxiety may be more likely to manifest through cultural and identity-based concerns rather than demands for additional economic redistribution. This institutional buffering effect suggests that the political consequences of technological change are not universal but instead depend critically on existing welfare state arrangements. This would be in line with findings that have shown that welfare state contexts play an important role in shaping policy preferences to technological change (Busemeyer and Tober, 2023).

Second, these results provide strong empirical support for theoretical arguments about the misattribution of automation-related economic hardship. The prominence of immigration attitudes in mediating automation’s political effects aligns with research suggesting that workers often fail to correctly identify technology as the source of their economic challenges, instead redirecting blame toward more visible and politically salient targets (Gallego and Kurer, 2022; Wu, 2022). This misattribution may be facilitated by political entrepreneurs who find it strategically advantageous to blame immigration and trade rather than technological progress, given that the latter offers fewer clear policy solutions and probably lacks the emotional resonance of cultural threat narratives. Thus, future research should not only examine the demand-side of politics but also look at the supply-side and explore how parties and politicians frame (AI-based) automation.

Third, the unexpected finding that immigration attitudes also influence populist left voting, though to a lesser extent than populist right voting, needs further theoretical and empirical consideration. This cross-

⁵One might think of Oesch’s occupational class schema (Oesch, 2006) as a more fine-grained option here, but the arguments I outlined above about why high skilled jobs should be reinstated fit the education proxy arguably better than Oesch’s class distinctions. Additionally, in the spirit of Occam’s razor, one should prefer the more parsimonious explanation, which in this case, considering the better fitting arguments, is the simple education proxy.

cutting effect may reflect a broader anti-establishment sentiment among automation-vulnerable workers, where negative views toward immigration become part of a generalized dissatisfaction with the political status quo rather than specific xenophobic ideology, from which then also populist left parties benefit. Alternatively, this pattern might indicate that some populist left parties have adopted more restrictive immigration positions to compete for working-class voters, blurring traditional left-right distinctions on cultural issues while maintaining redistributive economic platforms. This latter approach could be observed in the last German national election, where the BSW party combined a strong redistributive agenda with harsh anti-immigration policies. While this approach wasn't entirely unsuccessful, it ultimately failed to propel the party past the 5% threshold into the German parliament.

Lastly, I want to discuss the direct practical implications these findings have for European policymaking. The EU seeks to accelerate AI development as it has been argued that it is lacking behind the USA and China in AI development and adoption. While the EU AI act from 2024 provides a first regulation of AI itself, there have already been voices that call for a revision that weakens the existing law. There have been almost no calls for accompanying social lawmaking that aims to buffer the distributional consequences of AI automation. Considering the evidence that automation vulnerability increases support for anti-establishment parties, this suggests that policies promoting AI innovation without addressing distributional consequences may inadvertently undermine political support for the very institutions needed to govern technological change. An approach that is more thought through might combine increased investment in AI research and development with proactive measures to support automation-vulnerable workers, such as portable benefits systems, retraining programs targeted at occupations that face displacement, and regional development policies that create alternative employment opportunities in areas most affected by technological displacement. Such an approach would recognize that the EU's competitive disadvantage in AI development relative to the US and China cannot be addressed through technological catch-up alone, but requires managing the social and political prerequisites for sustained innovation.

These findings ultimately speak to a fundamental challenge facing contemporary democracies: how to manage technological progress in ways that strengthen rather than undermine democratic institutions. The tendency for automation-vulnerable workers to redirect economic anxieties toward immigration rather than technology itself suggests that democratic societies may be poorly equipped to have rational public debates about the costs and benefits of technological change. If voters systematically misattribute technology-driven disruption to more politically salient targets, democratic accountability for technology policy becomes nearly impossible. Nonetheless, the political consequences of technological change are not predetermined, but are shaped by how societies choose to manage technological transitions. The stakes of getting this right are considerable: AI's transformative potential for economic growth can only be realized if democratic institutions can maintain legitimacy and effectiveness throughout the transition. The alternative, a future where technological progress and democratic governance become increasingly incompatible, represents a challenge that extends far beyond any single policy domain.

References

- Acemoglu, D. (2024). The Simple Macroeconomics of AI. <https://doi.org/10.3386/w32487>
- Acemoglu, D., & Restrepo, P. Artificial Intelligence, Automation, and Work. In: In *The Economics of Artificial Intelligence: An Agenda*. University of Chicago Press, 2018, pp. 197–236. ISBN: 978-0-226-61333-8.
- Acemoglu, D., & Restrepo, P. (2019). Automation and New Tasks: How Technology Displaces and Reinstates Labor. <https://doi.org/10.3386/w25684>
- Acemoglu, D., & Restrepo, P. (2022). Tasks, Automation, and the Rise in U.S. Wage Inequality. *Econometrica*, 90(5), 1973–2016. <https://doi.org/10.3982/ECTA19815>
- Agrawal, A., Gans, J. S., & Goldfarb, A. (2019). Artificial Intelligence: The Ambiguous Labor Market Impact of Automating Prediction. *Journal of Economic Perspectives*, 33(2), 31–50. <https://doi.org/10.1257/jep.33.2.31>
- Akkerman, T. (2015). Gender and the radical right in western europe: A comparative analysis of policy agendas. *Patterns of Prejudice*, 49(1-2), 37–60. <https://doi.org/10.1080/0031322X.2015.1023655>
- Alekseeva, L., Azar, J., Giné, M., Samila, S., & Taska, B. (2021). The demand for AI skills in the labor market. *Labour Economics*, 71, 102002. <https://doi.org/10.1016/j.labeco.2021.102002>
- Alesina, A., & Giuliano, P. Chapter 4 - Preferences for Redistribution (J. Benhabib, A. Bisin, & M. O. Jackson, Eds.). In: *Handbook of Social Economics* (J. Benhabib, A. Bisin, & M. O. Jackson, Eds.). Ed. by Benhabib, J., Bisin, A., & Jackson, M. O. Vol. 1. North-Holland, 2011, pp. 93–131. <https://doi.org/10.1016/B978-0-444-53187-2.00004-8>
- Algan, Y., Guriev, S., Papaioannou, E., & Passari, E. (2017). The European Trust Crisis and the Rise of Populism. *Brookings Papers on Economic Activity*, 48(2 (Fall)), 309–400. <https://doi.org/10.1353/eca.2017.0015>
- Anelli, M., Colantone, I., & Stanig, P. (2019). We Were the Robots: Automation and Voting Behavior in Western Europe. <https://doi.org/10.2139/ssrn.3427624>
- Anelli, M., Colantone, I., & Stanig, P. (2021). Individual vulnerability to industrial robot adoption increases support for the radical right. *Proceedings of the National Academy of Sciences*, 118(47). <https://doi.org/10.1073/pnas.2111611118>
- Autor, D. H. (2015). Why Are There Still So Many Jobs? The History and Future of Workplace Automation. *Journal of Economic Perspectives*, 29(3), 3–30. <https://doi.org/10.1257/jep.29.3.3>
- Autor, D. H., Chin, C., Salomons, A., & Seegmiller, B. (2024). New Frontiers: The Origins and Content of New Work, 1940–2018. *The Quarterly Journal of Economics*, 139(3), 1399–1465. <https://doi.org/10.1093/qje/qjae008>
- Autor, D. H., & Dorn, D. (2013). The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market. *American Economic Review*, 103(5), 1553–1597. <https://doi.org/10.1257/aer.103.5.1553>
- Bandau, F. (2023). What Explains the Electoral Crisis of Social Democracy? A Systematic Review of the Literature. *Government and Opposition*, 58(1), 183–205. <https://doi.org/10.1017/gov.2022.10>
- Banks, A. J., & Valentino, N. A. (2012). Emotional Substrates of White Racial Attitudes. *American Journal of Political Science*, 56(2), 286–297. <https://doi.org/10.1111/j.1540-5907.2011.00561.x>
- Bauer, M., Cahliková, J., Chytilová, J., Roland, G., & Želinský, T. (2021). Shifting Punishment on Minorities: Experimental Evidence of Scapegoating. <https://doi.org/10.2139/ssrn.3906623>
- Bonikowski, B., & Stuhler, O. (2022). Reclaiming the Past to Transcend the Present: Nostalgic Appeals in U.S. Presidential Elections. *Sociological Forum*, 37(S1), 1263–1293. <https://doi.org/10.1111/socf.12838>

- Borwein, S., B., Bart, L., Peter John, L.-W., Blake, & and Magistro, B. (2025). Perceived technological threat and vote choice: Evidence from 15 European democracies. *West European Politics*, 48(3), 534–561. <https://doi.org/10.1080/01402382.2023.2297601>
- Brader, T., & Marcus, G. E. Emotion and political psychology. In: In *The Oxford handbook of political psychology*, 2nd ed. New York, NY, US: Oxford University Press, 2013, pp. 165–204.
- Bresnahan, T. Chapter 18 - General Purpose Technologies (B. H. Hall & N. Rosenberg, Eds.). In: *Handbook of the Economics of Innovation* (B. H. Hall & N. Rosenberg, Eds.). Ed. by Hall, B. H., & Rosenberg, N. Vol. 2. Handbook of the Economics of Innovation. North-Holland, 2010, pp. 761–791. [https://doi.org/10.1016/S0169-7218\(10\)02002-2](https://doi.org/10.1016/S0169-7218(10)02002-2)
- Brynjolfsson, E., & McAfee, A. (2017). Artificial intelligence, for real. *Harvard business review*, 1(1), 1–31.
- Brynjolfsson, E., Mitchell, T., & Rock, D. (2018). What Can Machines Learn and What Does It Mean for Occupations and the Economy? *AEA Papers and Proceedings*, 108, 43–47. <https://doi.org/10.1257/pandp.20181019>
- Brynjolfsson, E., Rock, D., & Syverson, C. (2021). The Productivity J-Curve: How Intangibles Complement General Purpose Technologies. *American Economic Journal: Macroeconomics*, 13(1), 333–372. <https://doi.org/10.1257/mac.20180386>
- Burgoon, B., van Noort, S., Rooduijn, M., & Underhill, G. (2019). Positional Deprivation and Support for Radical Right and Radical Left Parties. *Economic Policy*, 34. <https://doi.org/10.1093/epolic/eiy017>
- Bursztyn, L., Egorov, G., Haaland, I., Rao, A., & Roth, C. (2022). Scapegoating during Crises. *AEA Papers and Proceedings*, 112, 151–155. <https://doi.org/10.1257/pandp.20221069>
- Busemeyer, M. R., & Tober, T. (2023). Dealing with Technological Change: Social Policy Preferences and Institutional Context. *Comparative Political Studies*, 56(7), 968–999. <https://doi.org/10.1177/00104140221139381>
- Butz, D. A., & Yogeewaran, K. (2011). A new threat in the air: Macroeconomic threat increases prejudice against Asian Americans. *Journal of Experimental Social Psychology*, 47(1), 22–27. <https://doi.org/10.1016/j.jesp.2010.07.014>
- Choudhury, P., Starr, E., & Agarwal, R. (2020). Machine learning and human capital complementarities: Experimental evidence on bias mitigation. *Strategic Management Journal*, 41(8), 1381–1411. <https://doi.org/10.1002/smj.3152>
- Cinelli, C., & Hazlett, C. (2020). Making Sense of Sensitivity: Extending Omitted Variable Bias. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 82(1), 39–67. <https://doi.org/10.1111/rssb.12348>
- Dalton, R. J. (2019). *Citizen Politics: Public Opinion and Political Parties in Advanced Industrial Democracies*. CQ Press.
- Dancygier, R. M., & Walter, S. Globalization, Labor Market Risks, and Class Cleavages (P. Beramendi, S. Häusermann, H. Kitschelt, & H. Kriesi, Eds.). In: *The Politics of Advanced Capitalism* (P. Beramendi, S. Häusermann, H. Kitschelt, & H. Kriesi, Eds.). Ed. by Beramendi, P., Häusermann, S., Kitschelt, H., & Kriesi, H. Cambridge: Cambridge University Press, 2015, pp. 133–56.
- Deming, D., Ong, C., & Summers, L. H. Technological Disruption in the US Labor Market (M. S. Kearney & L. Pardue, Eds.). In: *Strengthening America's Economic Dynamism* (M. S. Kearney & L. Pardue, Eds.). Ed. by Kearney, M. S., & Pardue, L. Washington, DC: Aspen Institute, 2024.
- Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702), 1306–1308. <https://doi.org/10.1126/science.adj0998>

- European Social Survey ERIC (ESS ERIC). (2009). European Social Survey (ESS), Round 4 - 2008. <https://doi.org/10.21338/NSD-ESS4-2008>
- European Social Survey ERIC (ESS ERIC). (2011). European Social Survey (ESS), Round 5 - 2010. <https://doi.org/10.21338/NSD-ESS5-2010>
- European Social Survey ERIC (ESS ERIC). (2013). European Social Survey (ESS), Round 6 - 2012. <https://doi.org/10.21338/NSD-ESS6-2012>
- European Social Survey ERIC (ESS ERIC). (2015). European Social Survey (ESS), Round 7 - 2014. <https://doi.org/10.21338/NSD-ESS7-2014>
- European Social Survey ERIC (ESS ERIC). (2017). European Social Survey (ESS), Round 8 - 2016. <https://doi.org/10.21338/NSD-ESS8-2016>
- European Social Survey ERIC (ESS ERIC). (2019). European Social Survey (ESS), Round 9 - 2018. <https://doi.org/10.21338/NSD-ESS9-2018>
- European Social Survey ERIC (ESS ERIC). (2022). European Social Survey (ESS), Round 10 - 2020. <https://doi.org/10.21338/NSD-ESS10-2020>
- European Social Survey European Research Infrastructure (ESS ERIC). (2024). ESS11 - integrated file, edition 2.0. <https://doi.org/10.21338/ESS11E02.0>
- Felten, E. W., Raj, M., & Seamans, R. (2019). The Occupational Impact of Artificial Intelligence: Labor, Skills, and Polarization. <https://doi.org/10.2139/ssrn.3368605>
- Fenoaltea, E. M., Mazzilli, D., Patelli, A., Sbardella, A., Tacchella, A., Zaccaria, A., Trombetti, M., & Pietronero, L. (2024). Follow the money: A startup-based measure of AI exposure across occupations, industries and regions. <https://doi.org/10.48550/arXiv.2412.04924>
- Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to computerisation? *Technological Forecasting and Social Change*, 114, 254–280. <https://doi.org/10.1016/j.techfore.2016.08.019>
- Gallego, A., Kuo, A., Manzano, D., & Fernández-Albertos, J. (2022). Technological Risk and Policy Preferences. *Comparative Political Studies*, 55(1), 60–92. <https://doi.org/10.1177/00104140211024290>
- Gallego, A., & Kurer, T. (2022). Automation, Digitalization, and Artificial Intelligence in the Workplace: Implications for Political Behavior. *Annual Review of Political Science*, 25(Volume 25, 2022), 463–484. <https://doi.org/10.1146/annurev-polisci-051120-104535>
- Gamez-Djokic, M., & Waytz, A. (2020). Concerns About Automation and Negative Sentiment Toward Immigration. *Psychological Science*, 31(8), 987–1000. <https://doi.org/10.1177/0956797620929977>
- Google. (2025). Google Trends for AI and ChatGPT. <https://trends.google.de/trends/explore?date=2022-03-05%202025-06-03&q=ChatGPT,AI&hl=de>
- Graetz, G., & Michaels, G. (2018). Robots at Work. *The Review of Economics and Statistics*, 100(5), 753–768. https://doi.org/10.1162/rest_a_00754
- Hall, P. A., & Evans, G. Representation Gaps: Changes in Popular Preferences and the Structure of Partisan Competition in the Developed Democracies. In: *Unpublished Manuscript*. Minda de Gunzburg Center for European Studies, Harvard University, 2019.
- Hemmer, P., Schemmer, M., Kühn, N., Vössing, M., & Satzger, G. (2024). Complementarity in human-ai collaboration: Concept, sources, and evidence. <https://arxiv.org/abs/2404.00029>
- Ho, D. E., Imai, K., King, G., & Stuart, E. A. (2007). Matching as Nonparametric Preprocessing for Reducing Model Dependence in Parametric Causal Inference. *Political Analysis*, 15(3), 199–236. <https://doi.org/10.1093/pan/mpl013>

- Hobolt, S. B. (2016). The Brexit vote: A divided nation, a divided continent. *Journal of European Public Policy*, 23(9), 1259–1277. <https://doi.org/10.1080/13501763.2016.1225785>
- Im, Z. J., Mayer, N., Palier, B., & Rovny, J. (2019). The “losers of automation”: A reservoir of votes for the radical right? *Research & Politics*, 6(1), 2053168018822395. <https://doi.org/10.1177/2053168018822395>
- Imai, K., Keele, L., & Tingley, D. (2010a). A general approach to causal mediation analysis. *Psychological Methods*, 15(4), 309–334. <https://doi.org/10.1037/a0020761>
- Imai, K., Keele, L., Tingley, D., & Yamamoto, T. (2011). Unpacking the Black Box of Causality: Learning about Causal Mechanisms from Experimental and Observational Studies. *American Political Science Review*, 105(4), 765–789. <https://doi.org/10.1017/S0003055411000414>
- Imai, K., Keele, L., & Yamamoto, T. (2010b). Identification, Inference and Sensitivity Analysis for Causal Mediation Effects. *Statistical Science*, 25(1), 51–71. <https://doi.org/10.1214/10-STS321>
- Inglehart, R., & Norris, P. (2000). The Developmental Theory of the Gender Gap: Women’s and Men’s Voting Behavior in Global Perspective. *International Political Science Review / Revue internationale de science politique*, 21(4), 441–463. <https://doi.org/10.1177/0192512100214007>
- Kahl, S. (2005). The Religious Roots of Modern Poverty Policy: Catholic, Lutheran, and Reformed Protestant Traditions Compared. *European Journal of Sociology / Archives Européennes de Sociologie*, 46(1), 91–126. <https://doi.org/10.1017/S0003975605000044>
- Keman, H. (2011). Third Ways and Social Democracy: The Right Way to Go? *British Journal of Political Science*, 41(3), 671–680. <https://doi.org/10.1017/S0007123410000475>
- Krosch, A. R., & Amodio, D. M. (2014). Economic scarcity alters the perception of race. *PNAS Proceedings of the National Academy of Sciences of the United States of America*, 111(25), 9079–9084. <https://doi.org/10.1073/pnas.1404448111>
- Kurer, T. (2020). The declining middle: Occupational change, social status, and the populist right. *Comparative Political Studies*, 53(10-11), 1798–1835. <https://doi.org/10.1177/0010414020912283>
- Kurer, T., & Häusermann, S. Automation Risk, Social Policy Preferences, and Political Participation (M. R. Busemeyer, A. Kemmerling, K. Van Kersbergen, & P. Marx, Eds.). In: *Digitalization and the Welfare State* (M. R. Busemeyer, A. Kemmerling, K. Van Kersbergen, & P. Marx, Eds.). Ed. by Busemeyer, M. R., Kemmerling, A., Van Kersbergen, K., & Marx, P. Oxford University Press, 2022, pp. 139–156. <https://doi.org/10.1093/oso/9780192848369.003.0008>
- Lin, J. (2011). Technological Adaptation, Cities, and New Work. *The Review of Economics and Statistics*, 93(2), 554–574. https://doi.org/10.1162/REST_a.00079
- Maxwell, R. (2019). Cosmopolitan Immigration Attitudes in Large European Cities: Contextual or Compositional Effects? *American Political Science Review*, 113(2), 456–474. <https://doi.org/10.1017/S0003055418000898>
- Meltzer, A. H., & Richard, S. F. (1981). A Rational Theory of the Size of Government. *Journal of Political Economy*, 89(5), 914–927. <https://doi.org/10.1086/261013>
- Milner, H. V. (2021). Voting for Populism in Europe: Globalization, Technological Change, and the Extreme Right. *Comparative Political Studies*, 54(13), 2286–2320. <https://doi.org/10.1177/0010414021997175>
- Mokyr, J. (1990). *The Lever of Riches: Technological Creativity and Economic Progress*. Oxford University Press.

- Oesch, D. (2006). Coming to grips with a changing class structure: An analysis of employment stratification in Britain, Germany, Sweden and Switzerland. *International Sociology*, 21(2), 263–288. <https://doi.org/10.1177/0268580906061379>
- Olmstead, A. L., & Rhode, P. W. (2001). Reshaping the landscape: The impact and diffusion of the tractor in American agriculture, 1910–1960. *The Journal of Economic History*, 61(3), 663–698. <https://doi.org/10.1017/S0022050701030042>
- Oskarson, M., & Demker, M. (2015). Room for Realignment: The Working-Class Sympathy for Sweden Democrats. *Government and Opposition*, 50(4), 629–651. <https://doi.org/10.1017/gov.2014.41>
- Persson, M. (2015). Education and Political Participation. *British Journal of Political Science*, 45(3), 689–703. <https://doi.org/10.1017/S0007123413000409>
- Petersen, M. B. (2010). Distinct Emotions, Distinct Domains: Anger, Anxiety and Perceptions of Intentionality. *The Journal of Politics*, 72(2), 357–365. <https://doi.org/10.1017/S002238160999079X>
- Peterson, P. E. (1981). *City Limits*. University of Chicago Press.
- Potrafke, N. (2009). Did globalization restrict partisan politics? An empirical evaluation of social expenditures in a panel of OECD countries. *Public Choice*, 140(1), 105–124. <https://doi.org/10.1007/s11127-009-9414-2>
- Rehm, P. (2009). Risks and Redistribution: An Individual-Level Analysis. *Comparative Political Studies*, 42(7), 855–881. <https://doi.org/10.1177/0010414008330595>
- Rhodes-Purdy, M., Navarre, R., & Utych, S. M. (2021). Populist Psychology: Economics, Culture, and Emotions. *The Journal of Politics*, 83(4), 1559–1572. <https://doi.org/10.1086/715168>
- Rockall, E. J., Tavares, M. M., & Pizzinelli, C. (2025). AI Adoption and Inequality. *IMF Working Papers*, 2025(068). <https://doi.org/10.5089/9798229006828.001.A001>
- Rodríguez-Pose, A. (2018). The revenge of the places that don’t matter (and what to do about it). *Cambridge Journal of Regions, Economy and Society*, 11(1), 189–209. <https://doi.org/10.1093/cjres/rsx024>
- Scala, D. J., & Johnson, K. M. (2017). Political Polarization along the Rural-Urban Continuum? The Geography of the Presidential Vote, 2000–2016. *The ANNALS of the American Academy of Political and Social Science*, 672(1), 162–184. <https://doi.org/10.1177/0002716217712696>
- Schafer, J. L., & Kang, J. (2008). Average causal effects from nonrandomized studies: A practical guide and simulated example. *Psychological Methods*, 13(4), 279–313. <https://doi.org/10.1037/a0014268>
- Scheepers, P., Gijsberts, M., & Hello, E. (2002). Religiosity and Prejudice against Ethnic Minorities in Europe: Cross-National Tests on a Controversial Relationship. *Review of Religious Research*, 43(3), 242–265. <https://doi.org/10.2307/3512331>
- Scheve, K., & Slaughter, M. J. (1999). Labor-Market Competition and Individual Preferences Over Immigration Policy.
- Schöll, N., & Kurer, T. (2024). How technological change affects regional voting patterns. *Political Science Research and Methods*, 12(1), 94–112. <https://doi.org/10.1017/psrm.2022.62>
- Steenvoorden, E., & and Harteveld, E. (2018). The appeal of nostalgia: The influence of societal pessimism on support for populist radical right parties. *West European Politics*, 41(1), 28–52. <https://doi.org/10.1080/01402382.2017.1334138>
- Stuart, E. A. (2010). Matching Methods for Causal Inference: A Review and a Look Forward. *Statistical Science*, 25(1), 1–21. <https://doi.org/10.1214/09-STS313>
- Stubager, R. (2008). Education effects on authoritarian-libertarian values: A question of socialization. *The British Journal of Sociology*, 59(2), 327–350. <https://doi.org/10.1111/j.1468-4446.2008.00196.x>

- Susskind, D. (2020). *A world without work: Technology, automation and how we should respond*. Penguin UK.
- Thewissen, S., & Rueda, D. (2019). Automation and the Welfare State: Technological Change as a Determinant of Redistribution Preferences. *Comparative Political Studies*, 52(2), 171–208. <https://doi.org/10.1177/0010414017740600>
- Tilley, J., & Evans, G. (2014). Ageing and generational effects on vote choice: Combining cross-sectional and panel data to estimate APC effects. *Electoral Studies*, 33, 19–27. <https://doi.org/10.1016/j.electstud.2013.06.007>
- Voigtländer, N., & Voth, H.-J. (2015). Nazi indoctrination and anti-Semitic beliefs in Germany. *Proceedings of the National Academy of Sciences*, 112(26), 7931–7936. <https://doi.org/10.1073/pnas.1414822112>
- Walter, S. (2010). Globalization and the Welfare State: Testing the Microfoundations of the Compensation Hypothesis. *International Studies Quarterly*, 54(2), 403–426. <https://doi.org/10.1111/j.1468-2478.2010.00593.x>
- Webb, M. (2020). The Impact of Artificial Intelligence on the Labor Market. *Working Paper*. <https://doi.org/10.2139/ssrn.3482150>
- Westfall, J., & Yarkoni, T. (2016). Statistically Controlling for Confounding Constructs Is Harder than You Think. *PLOS ONE*, 11(3). <https://doi.org/10.1371/journal.pone.0152719>
- Wu, N. (2022). Misattributed blame? Attitudes toward globalization in the age of automation. *Political Science Research and Methods*, 10(3), 470–487. <https://doi.org/10.1017/psrm.2021.43>

A Appendix 1

Table A1: Party Classification by Country

Country	Party	Classification
Austria	Grüne	Mainstream Left
	KPÖ	Mainstream Left
	SPÖ	Mainstream Left
	NEOS	Mainstream Right
	ÖVP	Mainstream Right
	FPÖ	Populist Right
Belgium	Groen	Mainstream Left
	Parti Socialiste (PS)	Mainstream Left
	Partij van de Arbeid (PVDA)	Mainstream Left
	Socialistische Partij Anders (SPA)	Mainstream Left
	Sozialistische Partei (SP)	Mainstream Left
	Christen-Democratisch en Vlaams (CDV)	Mainstream Right
	Christlich Soziale Partei (CSP)	Mainstream Right
	DéFI (ex-FDF)	Mainstream Right
	Mouvement Réformateur (MR)	Mainstream Right
	Nieuw-Vlaamse Alliantie (NVA)	Mainstream Right
	Open Vlaamse Liberalen en Democraten (Open vld)	Mainstream Right
	Parti populaire (PP)	Populist Right
	Vlaams Belang (VB)	Populist Right
Denmark	Alternativet	Mainstream Left
	Enhedslisten – De Rød-Grønne	Mainstream Left
	Nye Borgerlige	Mainstream Left
	SF – Socialistisk Folkeparti	Mainstream Left
	Socialdemokratiet	Mainstream Left
	Konservative	Mainstream Right
	Liberal Alliance	Mainstream Right
	Venstre, Danmarks Liberale Parti	Mainstream Right
	Dansk Folkeparti	Populist Right
Finland	Puolue	Mainstream Left
	Sosiaalidemokraattinen	Mainstream Left
	Vasemmistoliitto	Mainstream Left
	Vihreä liitto	Mainstream Left
	Kansallinen Kokoomus	Mainstream Right
	Suomen Keskusta	Mainstream Right
	Suomen Kristillisdemokraatit	Mainstream Right
	Perussuomalaiset	Populist Right

Table A1: Party Classification by Country (*continued*)

Country	Party	Classification
France	Europe Écologie Les Verts (EELV)	Mainstream Left
	Génération.s	Mainstream Left
	Parti communiste français (PCF)	Mainstream Left
	Parti socialiste (PS)	Mainstream Left
	La république en Marche (LREM)	Mainstream Right
	Les républicains (LR)	Mainstream Right
	Mouvement démocrate (Modem)	Mainstream Right
	Union des démocrates et indépendants (UDI)	Mainstream Right
	Union populaire républicaine (UPR)	Mainstream Right
	La France insoumise (FI)	Populist Left
	Debout la France	Populist Right
	Rassemblement National (RN)	Populist Right
Germany	B90/Die Grünen	Mainstream Left
	SPD	Mainstream Left
	CDU	Mainstream Right
	CSU	Mainstream Right
	FDP	Mainstream Right
	FW	Mainstream Right
	Die Linke	Populist Left
	AFD	Populist Right
Greece	Communist Party of Greece	Mainstream Left
	Course of Freedom	Mainstream Left
	Movement for Change	Mainstream Left
	New Democracy	Mainstream Right
	MeRA25	Populist Left
	Syriza	Populist Left
	Golden Dawn	Populist Right
	Greek Solution	Populist Right
Ireland	AAA Solidarity–People Before Profit	Mainstream Left
	Green Party	Mainstream Left
	Labour Party	Mainstream Left
	Social Democrats	Mainstream Left
	Fianna Fáil	Mainstream Right
	Fine Gael	Mainstream Right
	Renua	Mainstream Right
	SF Sinn Féin	Populist Left

Table A1: Party Classification by Country (*continued*)

Country	Party	Classification
Italy	+Europa	Mainstream Left
	Liberi e Uguali	Mainstream Left
	Partito Comunista	Mainstream Left
	Partito Democratico	Mainstream Left
	Noi con l'Italia - UDC	Mainstream Right
	Potere al Popolo!	Populist Left
	Forza Italia	Populist Right
	Fratelli d'Italia	Populist Right
	Lega	Populist Right
Netherlands	GROENLINKS (GL)	Mainstream Left
	Partij van de Arbeid (P.v.d.A.)	Mainstream Left
	Christen-Democratisch Appèl (CDA)	Mainstream Right
	Volkspartij voor Vrijheid en Democratie (VVD)	Mainstream Right
	Socialistische Partij (SP)	Populist Left
	Partij voor de Vrijheid (PVV)	Populist Right
Norway	Arbeiderpartiet	Mainstream Left
	Miljøpartiet De Grønne	Mainstream Left
	Rødt	Mainstream Left
	Sosialistisk Venstreparti	Mainstream Left
	Høyre	Mainstream Right
	Kristelig Folkeparti	Mainstream Right
	Venstre	Mainstream Right
	Fremskrittspartiet	Populist Right
Portugal	Bloco de Esquerda	Mainstream Left
	CDU – Coligação Democrática Unitária	Mainstream Left
	LIVRE	Mainstream Left
	PESSOAS-ANIMAIS-	Mainstream Left
	Partido Comunista dos Trabalhadores Portugueses	Mainstream Left
	Partido Socialista	Mainstream Left
	Aliança	Mainstream Right
	CDS - Partido Popular	Mainstream Right
	Iniciativa Liberal	Mainstream Right
	Nós, Cidadãos!	Mainstream Right
	Partido Social Democrata	Mainstream Right
	Partido da Terra	Mainstream Right
	CHEGA	Populist Right
	Renovador	Populist Right

Table A1: Party Classification by Country (*continued*)

Country	Party	Classification
Spain	Candidatura d'Unitat	Mainstream Left
	Esquerra Republicana de Catalunya-Sobiranistes	Mainstream Left
	Euskal Herria Bildu	Mainstream Left
	Más País-Equo	Mainstream Left
	Partido Socialista Obrero Español (PSOE)	Mainstream Left
	Canarias	Mainstream Right
	Ciudadanos (C's)	Mainstream Right
	Coalición Canaria-Nueva	Mainstream Right
	Euzko Alderdi Jeltzalea-Partido Nacionalista Vasco	Mainstream Right
	Junts per Catalunya	Mainstream Right
	Partido Popular (PP)	Mainstream Right
	En Comú Podem	Populist Left
	Unidas Podemos	Populist Left
	Vox	Populist Right
Sweden	Feministiskt Initiativ	Mainstream Left
	Miljöpartiet	Mainstream Left
	Socialdemokraterna	Mainstream Left
	Vänsterpartiet	Mainstream Left
	Centerpartiet	Mainstream Right
	Kristdemokraterna	Mainstream Right
	Liberalerna	Mainstream Right
	Moderaterna	Mainstream Right
	Sverigedemokraterna	Populist Right
United Kingdom	Green Party of England and Wales	Mainstream Left
	Labour	Mainstream Left
	Liberals Democrats	Mainstream Left
	Plaid Cymru	Mainstream Left
	SDLP	Mainstream Left
	Scottish National Party	Mainstream Left
	Conservative	Mainstream Right
	Dup	Mainstream Right
	Sinn Féin	Populist Left
	Brexit Party	Populist Right
	UKIP	Populist Right

Note:

Source: Borwein et al. (2025)

Table A2: Variables from European Social Survey (ESS)

ESS Variable	Label	Question	Scale/Coding
gincdif	Redistribution Preferences	Using this card, please say to what extent you agree or disagree with each of the following statements. The government should take measures to reduce differences in income levels	1 Agree strongly, 2 Agree, 3 Neither agree nor disagree, 4 Disagree, 5 Disagree strongly
imbgeco	Immigration Attitudes	Would you say it is generally bad or good for [country]'s economy that people come to live here from other countries?	0 Bad for the economy, 1-9 Scale, 10 Good for the economy
prtcl	Party Identification	Is there a particular political party you feel closer to than all the other parties? Which one?	Country-specific party codes
lrscale	Left-Right Scale	In politics people sometimes talk of 'left' and 'right'. Using this card, where would you place yourself on this scale, where 0 means the left and 10 means the right?	0 Left, 1-9 Scale, 10 Right
polintr	Political Interest	How interested would you say you are in politics - are you...	1 Very interested, 2 Quite interested, 3 Hardly interested, 4 Not at all interested
facntr	Father's Birth Country	Was your father born in [country]?	1 Yes, 2 No
mocntr	Mother's Birth Country	Was your mother born in [country]?	1 Yes, 2 No
rlgdgr	Religiosity	Regardless of whether you belong to a particular religion, how religious would you say you are?	0 Not at all religious, 1-9 Scale, 10 Very religious
gndr	Gender	Gender	1 Male, 2 Female
agea	Age	Age calculated from birth year	Continuous age variable
eiscdf	Education Level	Education level	1 Less than lower secondary, 2 Lower secondary, 3 Lower tier upper secondary, 4 Upper tier upper secondary, 5 Advanced vocational, 6 Lower tertiary (BA), 7 Higher tertiary (MA+)
eiscdf	Father's Education	Father's education level	1 Less than lower secondary, 2 Lower secondary, 3 Lower tier upper secondary, 4 Upper tier upper secondary, 5 Advanced vocational, 6 Lower tertiary (BA), 7 Higher tertiary (MA+)

Table A2: Variables from European Social Survey (ESS) (*continued*)

ESS Variable	Label	Question	Scale/Coding
eiscedm	Mother's Education	Mother's education level	1 Less than lower secondary, 2 Lower secondary, 3 Lower tier upper secondary, 4 Upper tier upper secondary, 5 Advanced vocational, 6 Lower tertiary (BA), 7 Higher tertiary (MA+)
isco08	Occupation	What is/was the name or title of your main job? In your main job, what kind of work do/did you do most of the time?	4-digit ISCO occupation codes
hinctnta	Income Decile	Using this card, please tell me which letter describes your household's total income, after tax and compulsory deductions, from all sources?	1 1st decile (lowest), 2 2nd decile, ..., 10 10th decile (highest)
mbtru	Union Membership	Are you or have you ever been a member of a trade union or similar organisation?	1 Yes currently, 2 Yes previously, 3 No

Note:

Missing values (Refusal, Don't know, No answer) recoded as NA.

B Appendix 2

Table A3: Propensity Score Matching Specifications and Performance

Specification	Method	Distance	Treatment	Caliper	N Matched	Max SMD	Assessment
Ultra-tight caliper	Nearest	GLM	Median	0.05	25,871	0.047	Excellent
Very tight caliper	Nearest	GLM	Median	0.075	26,087	0.070	Excellent
Tight caliper (baseline)	Nearest	GLM	Median	0.10	26,297	0.082	Very Good
Tight GBM	Nearest	GBM	Median	0.10	23,562	0.080	Very Good
Alternative treatment	Nearest	GLM	Quartile	0.10	46,713	0.084	Very Good
1:2 matching ratio	Nearest	GLM	Median	0.10	26,297	0.084	Very Good
Neural network distance	Nearest	Neural Net	Median	0.10	23,569	0.103	Good
Standard caliper	Nearest	GLM	Median	0.20	27,031	0.173	Good
Random forest distance	Nearest	Random Forest	Median	0.10	23,721	0.297	Poor
Subclassification	Subclass	GLM	Median	None	120,028	0.203	Poor
Mahalanobis distance	Nearest	Mahalanobis	Median	None	31,085	0.862	Poor
No caliper restriction	Nearest	GLM	Median	None	31,085	1.640	Poor

Note:

Max SMD = Maximum Standardized Mean Difference. Treatment definitions:

Median = AI Score 1 vs 4; Quartile = AI Scores 1,2 vs 3,4.

Selected specification (Ultra-tight) highlighted in bold.

Table A4: Complete Covariate Balance Diagnostics for Ultra-Tight Matching Specification

Covariate	Unmatched Sample			Matched Sample			Balance Improvement (%)			
	Treated Mean	Control Mean	SMD	Treated Mean	Control Mean	SMD	SMD	Var.Ratio	eCDF Mean	eCDF Max
Matching Covariates										
Distance	0.792	0.595	1.261	0.661	0.653	0.047	96.3	95.6	95.7	94.8
Immigration Background	0.918	0.891	0.098	0.895	0.904	-0.032	67.2	—	67.2	67.2
Gender (Female)	0.511	0.511	-0.000	0.477	0.504	-0.055	-21423.4	—	-21423.4	-21423.4
Age	50.226	46.873	0.169	45.848	46.991	-0.058	65.9	13.0	40.3	-16.7
Parental Education	2.523	3.851	-0.896	3.464	3.466	-0.001	99.9	87.2	96.9	96.0
Income Decile	6.064	4.017	0.768	4.475	4.371	0.039	94.9	97.4	94.9	92.8
Country Indicators										
Austria	0.029	0.016	0.079	0.016	0.016	0.000	100.0	—	100.0	100.0
Belgium	0.043	0.072	-0.144	0.066	0.066	0.000	100.0	—	100.0	100.0
Bulgaria	0.039	0.037	0.006	0.037	0.037	0.000	100.0	—	100.0	100.0
Switzerland	0.041	0.034	0.035	0.036	0.036	0.000	100.0	—	100.0	100.0
Cyprus	0.012	0.012	0.003	0.011	0.011	0.000	100.0	—	100.0	100.0
Czech Republic	0.055	0.023	0.140	0.026	0.026	0.000	100.0	—	100.0	100.0
Germany	0.062	0.064	-0.007	0.070	0.070	0.000	100.0	—	100.0	100.0
Denmark	0.025	0.039	-0.089	0.037	0.037	0.000	100.0	—	100.0	100.0
Estonia	0.029	0.044	-0.093	0.043	0.043	0.000	100.0	—	100.0	100.0
Spain	0.046	0.040	0.027	0.042	0.042	0.000	100.0	—	100.0	100.0
Finland	0.041	0.056	-0.076	0.055	0.055	0.000	100.0	—	100.0	100.0
France	0.051	0.045	0.026	0.045	0.045	0.000	100.0	—	100.0	100.0
United Kingdom	0.030	0.043	-0.074	0.040	0.040	0.000	100.0	—	100.0	100.0
Greece	0.019	0.011	0.059	0.012	0.012	0.000	100.0	—	100.0	100.0
Croatia	0.033	0.015	0.101	0.017	0.017	0.000	100.0	—	100.0	100.0
Hungary	0.042	0.018	0.118	0.019	0.019	0.000	100.0	—	100.0	100.0
Ireland	0.049	0.046	0.012	0.044	0.044	0.000	100.0	—	100.0	100.0
Iceland	0.011	0.024	-0.118	0.022	0.022	0.000	100.0	—	100.0	100.0
Italy	0.035	0.016	0.102	0.018	0.018	0.000	100.0	—	100.0	100.0
Lithuania	0.031	0.041	-0.061	0.043	0.043	0.000	100.0	—	100.0	100.0
Latvia	0.012	0.014	-0.021	0.015	0.015	0.000	100.0	—	100.0	100.0
Montenegro	0.016	0.007	0.069	0.008	0.008	0.000	100.0	—	100.0	100.0
Netherlands	0.040	0.068	-0.143	0.063	0.063	0.000	100.0	—	100.0	100.0
Norway	0.039	0.072	-0.175	0.070	0.070	0.000	100.0	—	100.0	100.0
Poland	0.041	0.036	0.026	0.037	0.037	0.000	100.0	—	100.0	100.0
Portugal	0.029	0.021	0.047	0.021	0.021	0.000	100.0	—	100.0	100.0
Serbia	0.011	0.006	0.046	0.006	0.006	0.000	100.0	—	100.0	100.0
Sweden	0.028	0.038	-0.060	0.037	0.037	0.000	100.0	—	100.0	100.0
Slovenia	0.040	0.027	0.063	0.031	0.031	0.000	100.0	—	100.0	100.0
Slovakia	0.025	0.015	0.064	0.016	0.016	0.000	100.0	—	100.0	100.0
Sample Composition										
All	88,943	31,085								
Matched	25,871	25,871								
Unmatched	63,072	5,214								
Discarded	0	0								

Specification: Nearest-neighbor propensity score matching with ultra-tight caliper (0.05), exact matching on country, 1:1 ratio with replacement.

SMD = Standardized Mean Difference.

Balance Improvement shows percentage reduction in imbalance across different metrics (SMD, Variance Ratio, eCDF).

eCDF Mean: Measures the average difference between the cumulative distribution functions of the treatment and control groups across all values of a variable.

eCDF Max: Measures the maximum difference between the cumulative distribution functions (this is essentially the Kolmogorov-Smirnov statistic).

Negative improvement percentages for gender reflect extremely small baseline imbalance magnified by matching adjustments.

All country indicators achieve perfect balance (SMD = 0.000) due to exact matching constraints.

Table A5: Mediation Sensitivity Analysis: Thresholds for Eliminating Effects

Mediation Pathway	ρ Threshold	$R_M^2 \times R_Y^2$ Threshold	$R_{M\sim}^2 \times R_{Y\sim}^2$ Threshold	Robustness Assessment
Theoretically Expected Pathways				
Immigration $\rightarrow$ Populist Right	-0.3	0.09	0.018	High
Redistribution $\rightarrow$ Populist Left	0.2	0.04	0.004	Moderate
Cross Pathways				
Redistribution $\rightarrow$ Populist Right	0.1	0.01	0.002	Low
Immigration $\rightarrow$ Populist Left	-0.2	0.04	0.004	Moderate

Interpretation: Values represent the strength of unmeasured confounding required to eliminate the mediation effect. $\rho = 0.1$ means unobserved confounders would need correlation of 0.1 between mediator and outcome residuals. R^2 thresholds show proportion of unexplained variance that confounders must account for. Higher thresholds indicate more robust effects. Higher absolute values indicate greater robustness to unmeasured confounding. The main theoretical pathways show greater robustness than cross pathways, supporting the proposed theoretical framework.

C Appendix 3

Table A6: Effect of Automation Risk on Political Attitudes (Unmatched Sample, Fixed Effects)

	<i>Dependent variable:</i>	
	Redistribution Preferences	Immigration Attitudes
	(1)	(2)
AI Score (1-4)	0.038*** (0.006)	-0.247*** (0.016)
Parental Education	-0.036*** (0.005)	0.148*** (0.012)
Left-Right Scale	-0.100*** (0.012)	-0.092*** (0.024)
Gender (Female)	0.098*** (0.009)	-0.178*** (0.038)
Age	0.001 (0.001)	0.001 (0.002)
Income Decile	0.039*** (0.004)	-0.079*** (0.007)
Union Member	0.109*** (0.012)	0.027 (0.046)
Country FE	Yes	Yes
Year FE	Yes	Yes
Clustered SE	Country	Country
Observations	186,322	183,889
R ²	0.147	0.139
Adjusted R ²	0.147	0.139
Residual Std. Error	0.941 (df = 186269)	2.232 (df = 183836)
F Statistic	617.482*** (df = 52; 186269)	571.402*** (df = 52; 183836)

Note:

*p<0.1; **p<0.05; ***p<0.01

Country-clustered standard errors in parentheses.

All models include two-way fixed effects for country and year.

Table A7: Linear Mixed Effects Models (Matched Sample, Random Effects)

	<i>Dependent variable:</i>	
	Redistribution Preferences	Immigration Attitudes
	(1)	(2)
Treatment	0.042*** (0.003)	-0.240*** (0.007)
Parental Education	-0.025*** (0.003)	0.115*** (0.007)
Left-Right Scale	-0.134*** (0.002)	-0.125*** (0.005)
Gender (Female)	0.115*** (0.009)	-0.231*** (0.020)
Age	0.001*** (0.0003)	0.003*** (0.001)
Income Decile	0.047*** (0.002)	-0.068*** (0.004)
Union Member	0.097*** (0.011)	-0.059** (0.024)
Country Random Effects	Yes	Yes
Year Random Effects	Yes	Yes
Model	Linear Mixed	Linear Mixed
Observations	45,227	45,227
Log Likelihood	-63,126.560	-98,160.810
Akaike Inf. Crit.	126,275.100	196,343.600
Bayesian Inf. Crit.	126,371.000	196,439.500

Note:

*p<0.1; **p<0.05; ***p<0.01

Standard errors in parentheses.

Random intercepts for country and year.

Table A8: Treatment Effect Stability Across ChatGPT Release

Outcome Variable	Difference (Post - Pre)	Std. Error	t-stat	p-value	95% Confidence Interval	
					Lower	Upper
Redistribution Preferences	-0.069	0.050	-1.370	0.171	-0.167	0.029
Immigration Attitudes	-0.123	0.112	-1.092	0.275	-0.343	0.097

Notes: Tests whether treatment effects changed following ChatGPT's release (November 2022). Pre-period: 2021-2022; Post-period: 2023-2024. Confidence intervals calculated using t-distribution. Neither difference is statistically significant, indicating treatment effects remained stable despite increased AI salience post-ChatGPT.